

trgn515_assign1_sub_github

ranzhu

2026-02-10

Task 1: Describe briefly the differences in read length, base quality and GC content per sequencing platform (10pts)

Read Length

- The x-axis represents read length in base pairs (log scale), and the y-axis indicates the number of reads.
- Illumina forward reads distribution form a narrow peak centered at approximately 150 bp.
- PacBio reads are mainly distributed in the 10-20kb range.
- Nanopore reads range from a few kilobases to over 100 kb.

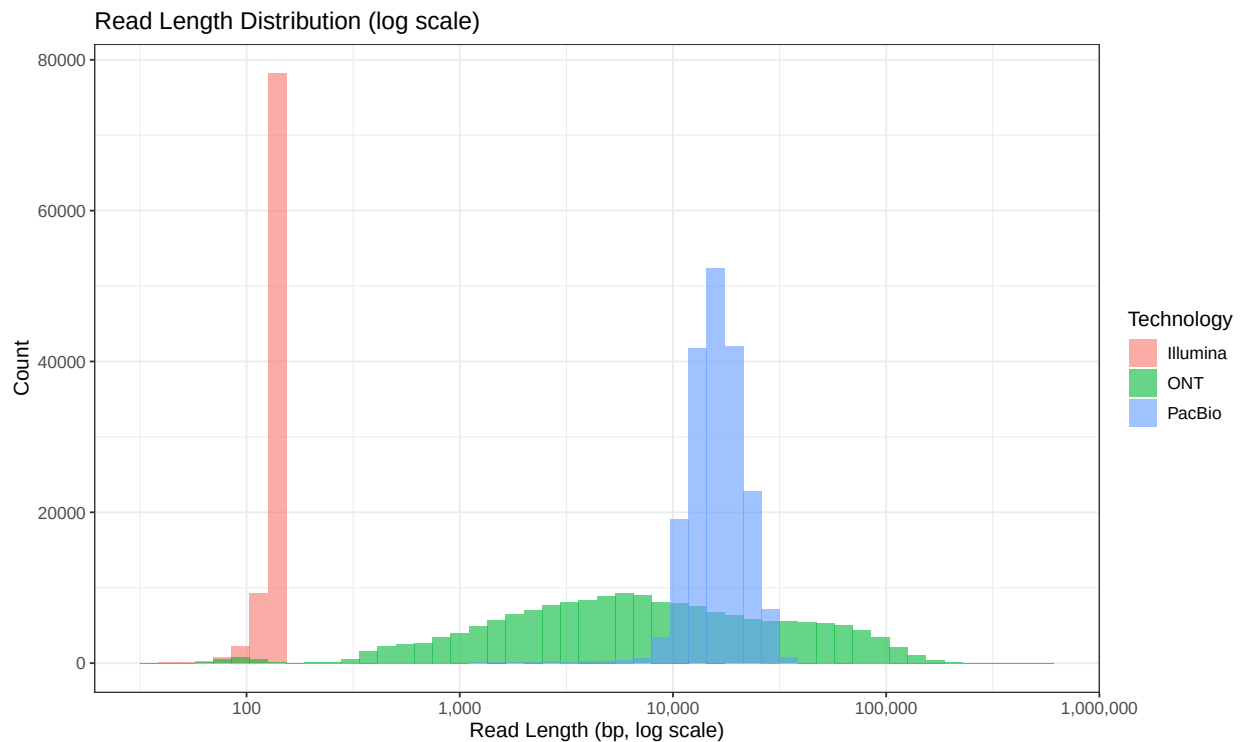


Figure 1: figure 1.shows read length distribution among Illumina, PacBio, and Nanopore.

Average Base Quality

- The x-axis represents the average base quality score per read, and the y-axis indicates the number of reads.
- Illumina reads show the highest average base quality, peaking around Q35–38, with read counts decreasing toward lower quality scores.
- PacBio reads shows intermediate quality scores, mainly between Q20 and Q35.
- Nanopore reads have lower average quality scores, primarily distributed around Q8–12.



Figure 2: figure 2.shows the distribution of average base quality scores for Illumina, PacBio, and ONT reads. The x-axis represents the average quality score per read, and the y-axis represents the number of reads (count) within each quality.

GC Content

- The x-axis represents GC content (%), and the y-axis indicates the number of reads.
- Illumina reads show a GC content distribution centered around 40–50%. * PacBio reads exhibit a similar distribution, also peaking around 40–50%.
- Nanopore reads display a slightly broader distribution spanning approximately 35–55%.

Extract raw read statistics (chr20) code

illumina forward

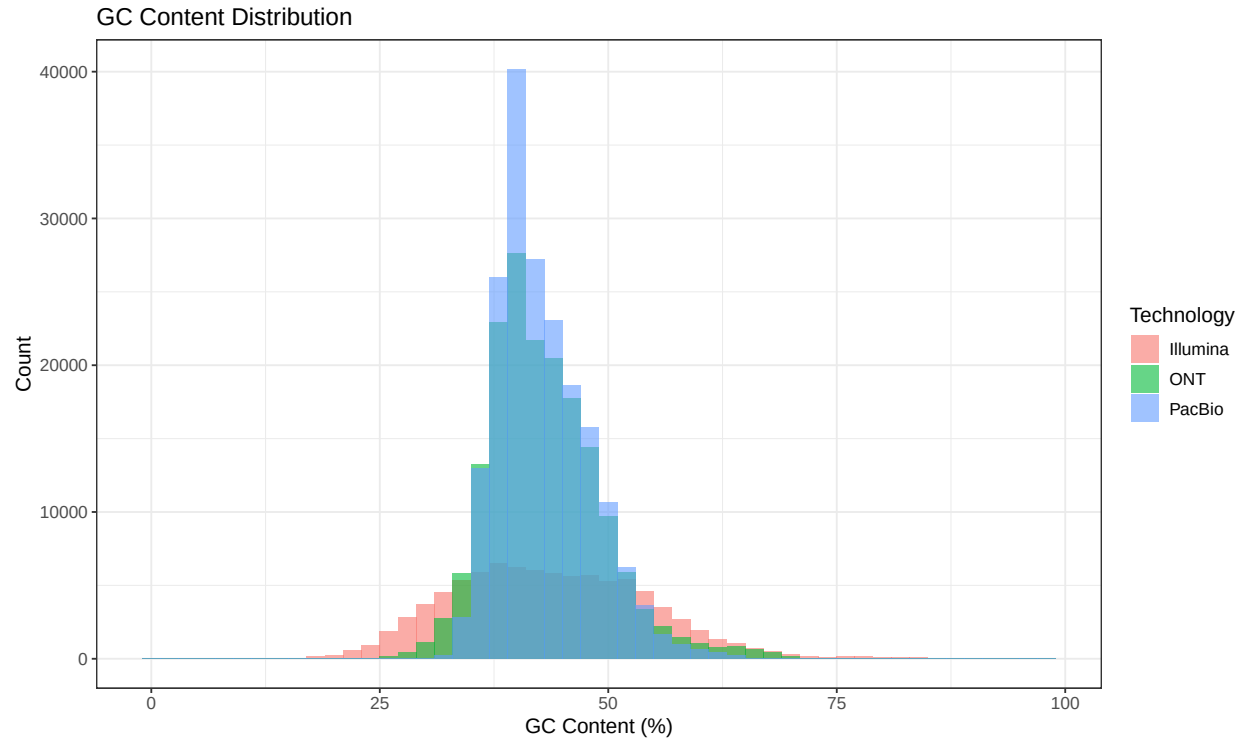


Figure 3: The x-axis shows the GC percentage calculated for each read, and the y-axis shows how many reads fall into each GC range.

```
seqtk sample -s 100 \
/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_illumina_chr20_1.fastq.gz \
200000 \
| seqkit fx2tab -n -l -q -g \
> /project2/msalomon_1816/ranzhu/assignment_1/question1_used_chr20/illumina_stats.txt
```

Pacbio

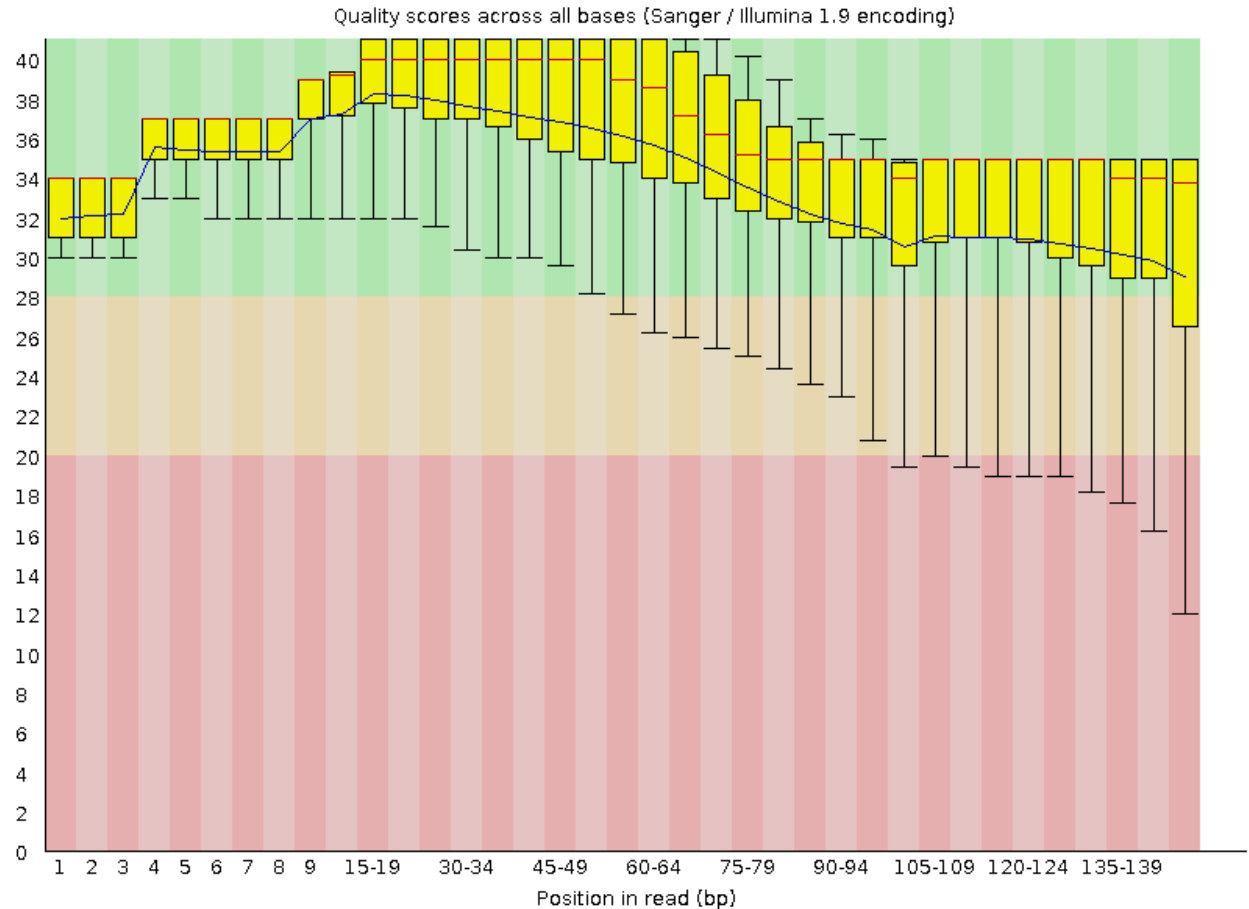
```
seqtk sample -s 100 \
/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_pacbio_chr20.fastq.gz \
200000 \
| seqkit fx2tab -n -l -q -g \
> /project2/msalomon_1816/ranzhu/assignment_1/question1_used_chr20/pacbio_stats.txt
```

Nanopore

```
seqtk sample -s 100 \
/project2/msalomon_1816/ranzhu/assignment_1/fastq/hg200_ont_chr20.fastq.gz \
200000 \
| seqkit fx2tab -n -l -q -g \
> /project2/msalomon_1816/ranzhu/assignment_1/question1_used_chr20/ont_stats.txt
```

Question 2: Run FastQc on Illumina Reads

Per base sequence quality



1. what is the plot representing?

- The plot shows the per-base quality scores across all reads from Illumina sequencing. The x-axis represents the base position along the read, and the y-axis indicates the Phred quality score. Each yellow boxplot represents the distribution of quality scores at a specific position in the reads, from position 1 to position 139.
- The background colors indicate quality zones: green (>28) for good quality, orange (20-28) for reasonable quality, and red (<20) for low quality.

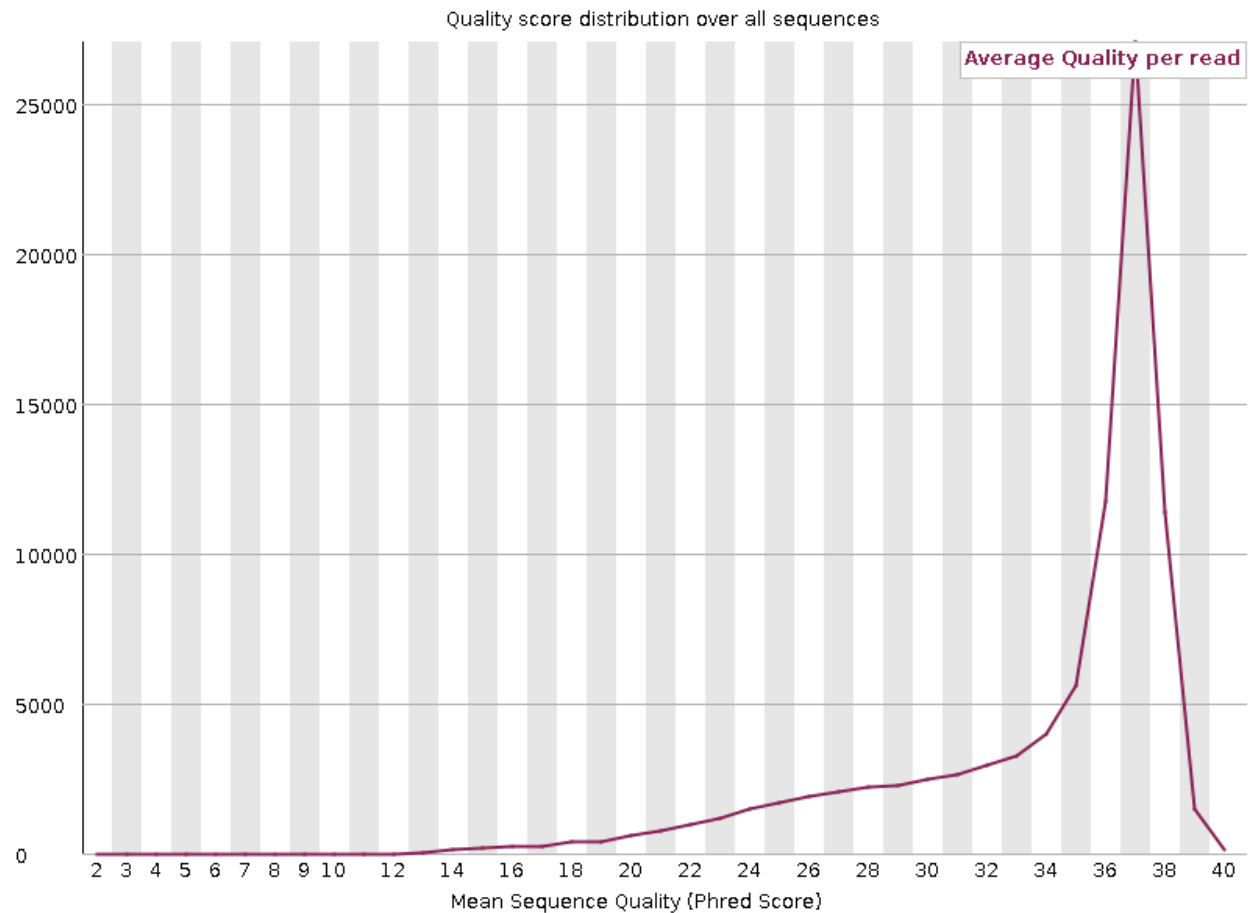
2. what quality issue (if any) is the plot showing?

- Quality scores remained above Q30 across most read positions, with a gradual decline toward the 3' prime end. The last ~20 bases (positions 120-139) shows lower median quality, and the minimum values reaching into the red zone as low as Q12-20.

3. How would you solve the issue (if any) represented in the plot?

- We can use Trimmomatic to trim low quality bases from the 3' end of reads.

Sequencing quality scores



1. what is the plot representing?

- The plot shows the distribution of average quality scores per read. The x axis shows the mean sequence quality ranging from 0 to 40, and the Y axis shows the count of reads with that average quality. Each point on the curve represents how many reads have that specific mean quality score across their entire length.
- The quality score distribution shows that most reads have very high average quality, with a peak around Q37–38 and a maximum count above 25,000. There are very few reads with an average quality below Q30, as the curve is close to zero in that range. Overall, this suggests that the sequencing data is high quality.

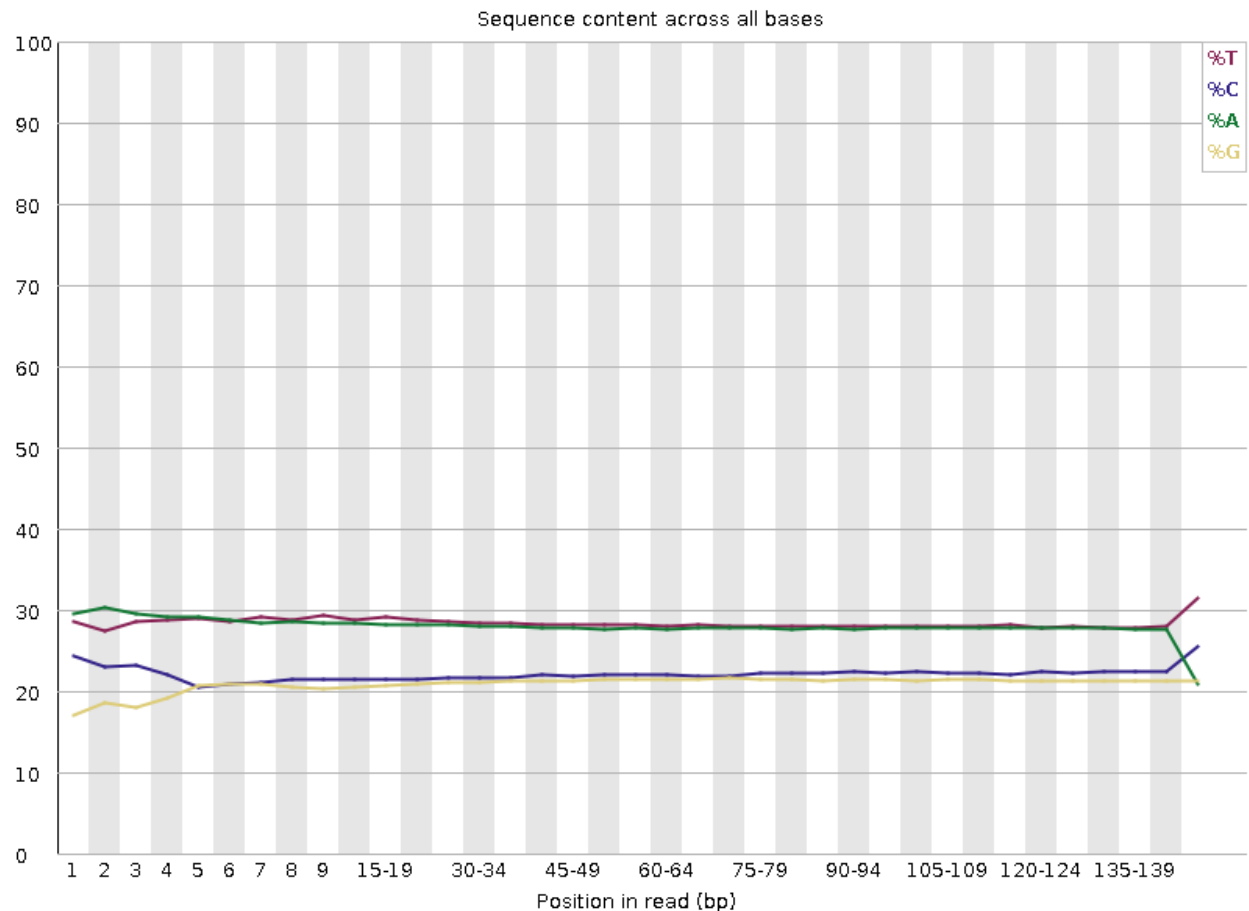
2. what quality issue (if any) is the plot showing?

- There is no significant quality issues for this plot.

3. How would you solve the issue (if any) represented in the plot?

- No action is needed in this case.

Per base sequence content



1. what is the plot representing?

- The plot shows the proportion of each nucleotide (A, T, C, and G) at each position across all reads. The x-axis represents the position along the read, and the y-axis represents the percentage of each base at that position.
- Slight variation in nucleotide proportions is observed at the first few base positions (1–5 bp) may be caused by library preparation. After approximately position 5, the proportions of A/T and C/G stabilize and remain relatively constant across the read. A and T each account for approximately 28–30%, while C and G each represent approximately 21–23%. Small variation are observed toward the end of the reads.

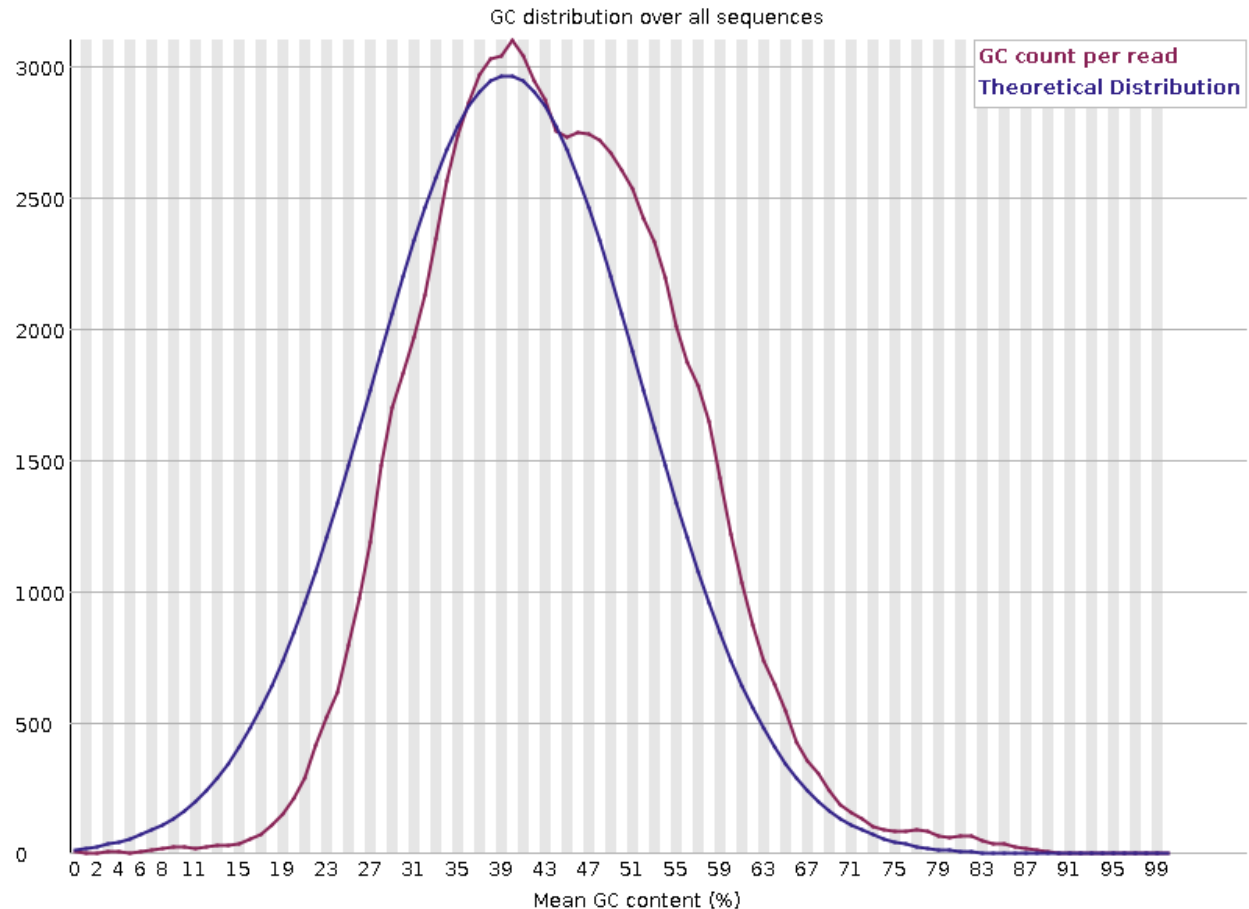
2. what quality issue (if any) is the plot showing?

- Slight variation is observed at positions 1–5 at the beginning of the reads, likely reflecting library preparation bias.

3. How would you solve the issue (if any) represented in the plot?

- If necessary, the first 5–10 bases could be trimmed.

Per sequence GC content



1. what is the plot representing?

- X axis shows the mean of GC content percentage, and y axis shows the length of sequence. The actual observed GC distribution (red line) is similar to the theoretical expected distribution (blue line), showing that the data are overall of good quality.

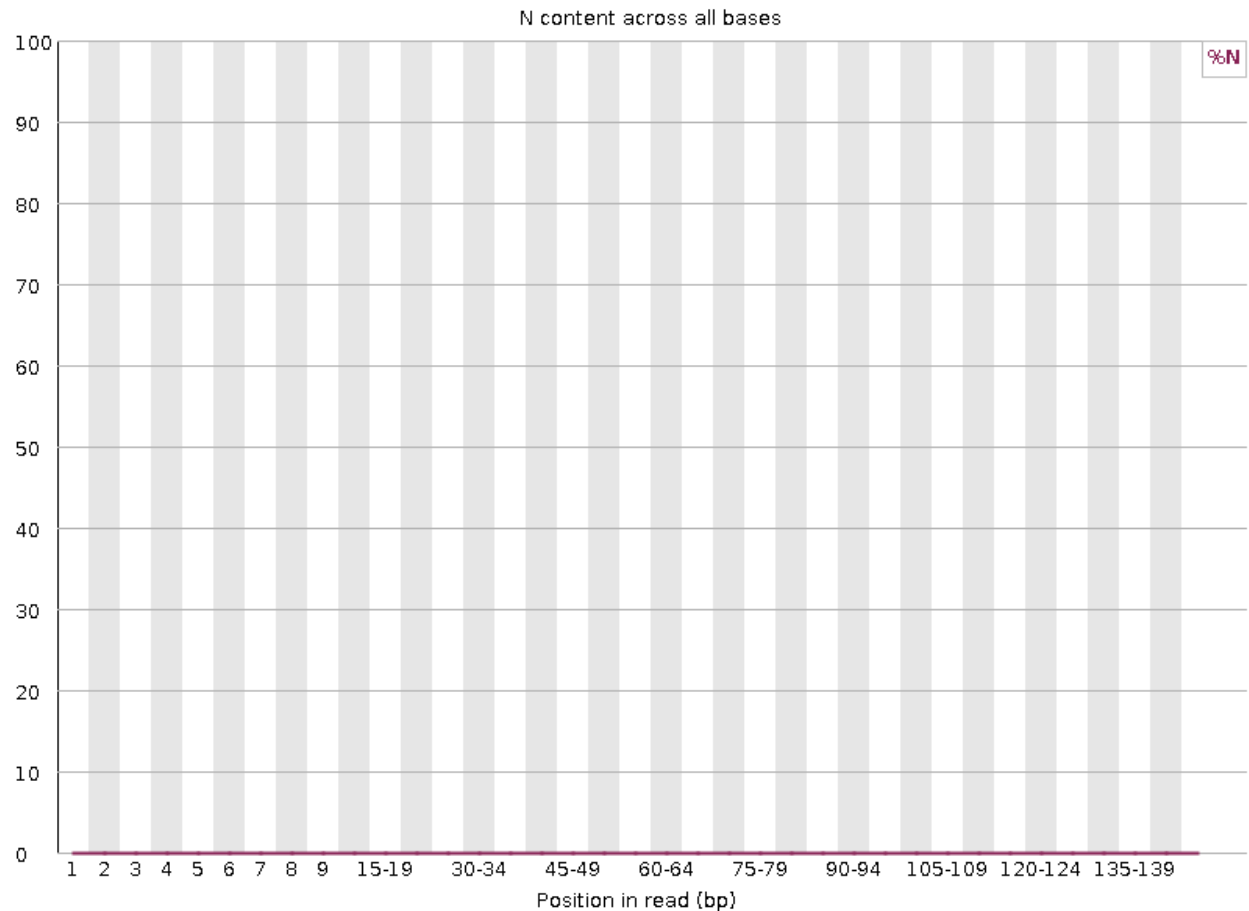
2. what quality issue (if any) is the plot showing?

- The GC distribution is close to the expected normal distribution. There is only a small difference between the observed and theoretical curves. This suggests that there is no serious GC-related quality problem in the dataset.

3. How would you solve the issue (if any) represented in the plot?

- No specific correction is needed.

Per base N content



1. what is the plot representing?

- This plot shows the percentage of unknown base calls (N) at each position across all Illumina forward reads. The x-axis represents the position in the read, while the y-axis indicates the percentage of bases called as N at each position.
- For every base position along the read length, the plot summarizes how often the sequencer was unable to confidently assign a nucleotide and instead reported an "N".
- The plot shows that the N content is close to 0% at all positions.

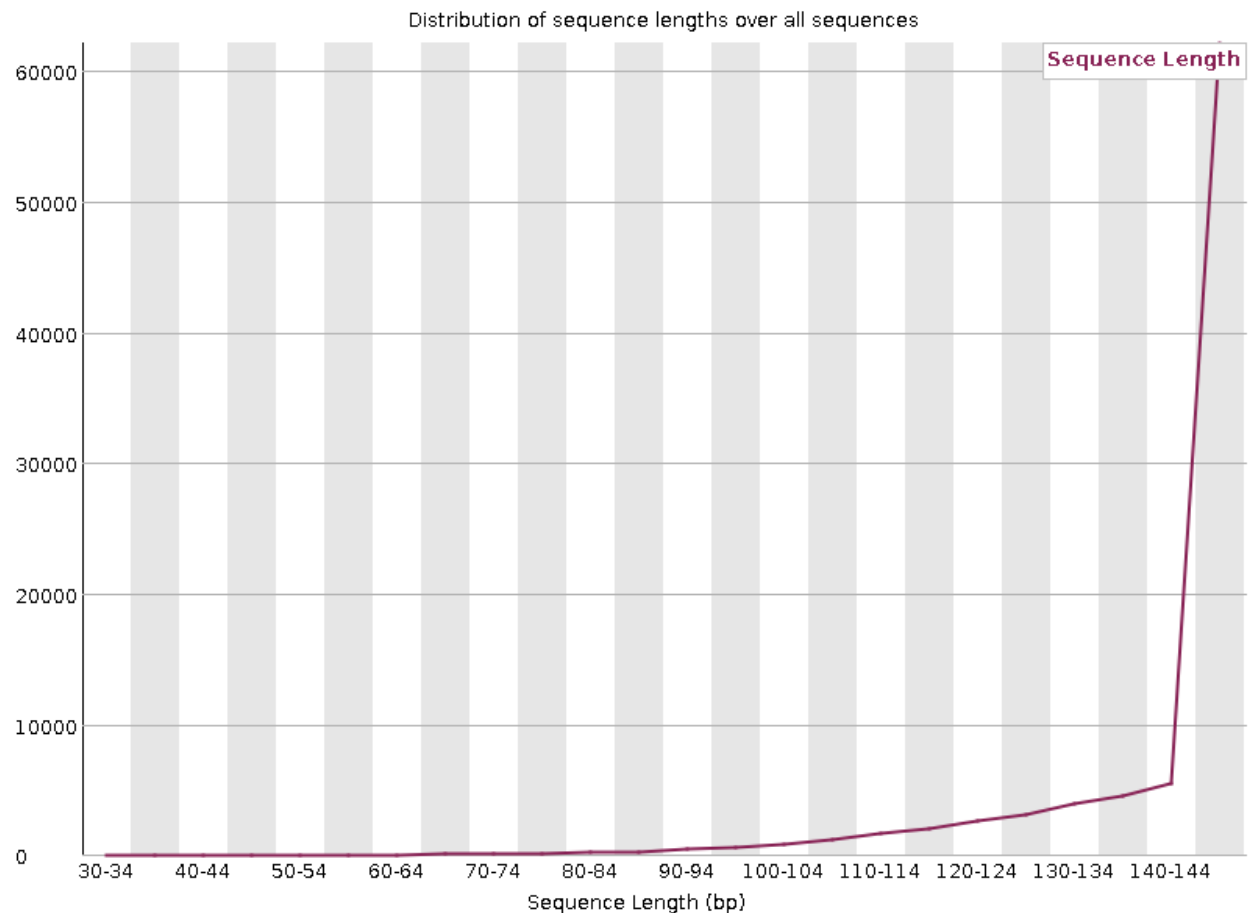
2. what quality issue (if any) is the plot showing?

- There is no quality issue for this plot.

3. How would you solve the issue (if any) represented in the plot?

- No corrective action is required.

Sequence Length Distribution



1. what is the plot representing?

- This plot shows the distribution of sequence lengths across all reads. The x-axis shows the sequence length in base pairs, and the y-axis shows the number of reads at each length. Each read is counted based on its final length after sequencing.
- Most reads are concentrated at around 140–144 bp. Only a small number of reads have shorter lengths.
- This is typically for illumina short reads because illumina sequencing runs for a fixed number of cycles, with each cycle incorporating and reading one nucleotide. Therefore, the theoretical read length is should be the same and determined by the number of sequencing cycles.

2. what quality issue (if any) is the plot showing?

- There is no issue with this plot.

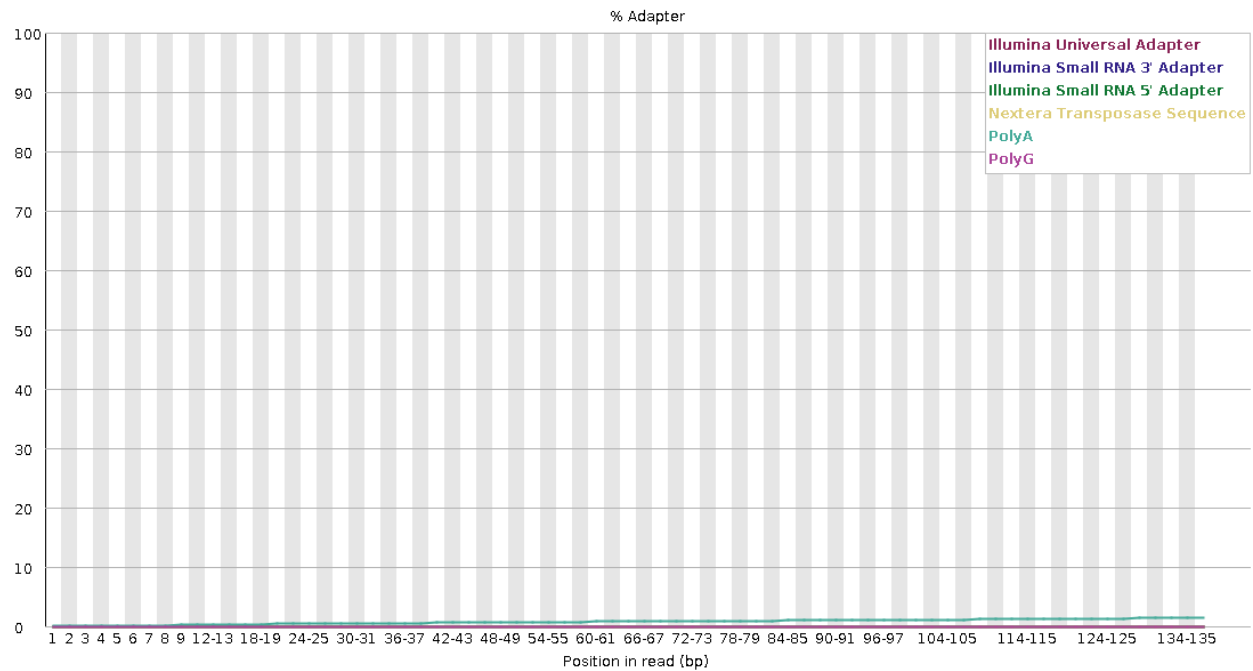
3. How would you solve the issue (if any) represented in the plot?

- no correction is needed in this case.

Overrepresented sequences

- No overrepresented sequences represented

Adapter Content



1. what is the plot representing?

- The plot shows the percentage of reads containing adapter-derived sequences at each base position across all Illumina forward reads. The x-axis represents the position in the read, while the y-axis represents the percentage of reads in which the adapter sequence is detected at that position.
- The plot does not indicate any significant adapter contamination. The adapter content remains near 0% across all positions.

2. what quality issue (if any) is the plot showing?

- The plot does not show any significant quality issue. Adapter content remains near 0% across all positions, indicating that there is no detectable adapter contamination in the reads.

3. How would you solve the issue (if any) represented in the plot?

- no correction is needed in this case.

Question 3: Run FastQc After Trimmomatic

Per base sequence quality

1. What has changed after running trimmomatic? (10pts)

*After trimming, the minimum quality scores at the 3' end increased, indicating that low-quality bases were removed.

2. Why did the change happen? (10pts)

- The change is mainly observed at the 3' end because trimming specifically removes low-quality bases from the ends of reads. After trimming, these poor-quality terminal bases were removed, so they no longer contribute to the FastQC statistics. The minimum quality values at the read ends increased slightly. Since most of the read already had high quality, those regions remained unchanged.

Per sequence quality scores

1. What has changed after running trimmomatic? (10pts)

- There is no significant change after running Trimmomatic. The overall distribution of per-sequence quality scores remains the same, with a similar peak and spread, indicating that trimming did not significantly affect the average quality of the reads.

2. Why did the change happen? (10pts)

- There is no significant change because trimming mainly removes a small number of low-quality bases at the ends of reads, which does not significantly affect the average quality per read.

Per base sequence content

1. What has changed after running trimmomatic? (10pts)

- In the before trimming data, the separation between A, T, C, and G is slightly more pronounced at positions 3–4 bp, with C and G showing a small deviation from A and T.
- In the after trimming data, the base composition curves at these positions are slightly closer together, resulting in a smoother base composition profile.

2. Why did the change happen? (10pts)

- The change occurred because trimming removed a small number of low-quality or short reads from the dataset.

Per sequence GC content

1. What has changed after running trimmomatic? (10pts)

- There is no substantial change in the per-sequence GC content after trimming.

2. Why did the change happen? (10pts)

- There is no change because Trimmomatic mainly removes low-quality bases at the ends of reads, which does not specifically affect GC- or AT-rich regions.

Per base N content

1. What has changed after running trimmomatic? (10pts)

- No change is observed in the per-base N content after trimming. In both the trimmed and untrimmed data, the N content remains at 0 percent across all positions, with no detectable peaks or increases along the reads.

2. Why did the change happen? (10pts)

- No change occurred because there were no ambiguous bases (N) present in the reads before trimming. Since Trimmomatic removes low-quality bases but does not introduce or specifically target N bases, the per-base N content remains unchanged after trimming.

Sequence Length Distribution

1. What has changed after running trimmomatic? (10pts)

- There is no substantial change in the sequence length distribution after trimming. Both plots show a dominant peak at full read length (~140–144 bp) and a small proportion of shorter reads. The overall distribution remains largely the same before and after trimming.

2. Why did the change happen? (10pts)

- There is no substantial change because most reads already had high quality and did not require trimming. Trimmomatic only affected a small fraction of reads, so the overall sequence length distribution remained largely unchanged.

Overrepresented sequences

- No overrepresented sequences

Adapter Content

1. What has changed after running trimmomatic? (10pts)

- There is no change observed after running trimmomatic.

2. Why did the change happen? (10pts)

- No change is observed because the sequencing data were already of high quality before trimming. Since there were no significant issues such as adapter contamination or low-quality bases, trimming did not have the impact on the dataset.

Question 4: Long Read QC

Pacbio

Metric	Before	After
Number of reads	191,592	180,511
Total bases (Gb)	3.20	3.04
Mean read length (bp)	16,687.5	16,826.4
Median read length (bp)	16,052.0	16,134.0
Read length N50 (bp)	17,394	17,440
Mean read quality	23.5	24.6
Median read quality	27.2	27.7
Reads > Q20 (%)	86.0	90.5
Reads > Q25 (%)	61.4	64.5

- From the statistics, the number of reads decreases slightly ($191,592 \rightarrow 180,511$), and total bases are modestly reduced ($3.20 \text{ Gb} \rightarrow 3.04 \text{ Gb}$), indicating that only a small fraction of reads was removed. Both mean and median read quality increase slightly ($23.5 \rightarrow 24.6$; $27.2 \rightarrow 27.7$), and the proportion of high-quality reads increases (reads >Q20: $86.0\% \rightarrow 90.5\%$; reads >Q25: $61.4\% \rightarrow 64.5\%$), demonstrating enrichment for higher-quality reads.
- There is slightly change for read length. The median read length increases marginally ($16,052 \text{ bp} \rightarrow 16,134 \text{ bp}$) and N50 changes minimally ($17,394 \text{ bp} \rightarrow 17,440 \text{ bp}$).
- From the plots, the read length histograms before and after filtering are very similar, with the main peak staying around 15–17 kb. The log-transformed histograms show a small reduction in the short-read tail, and the yield-by-length plots show a slight decrease in total yield.
- Overall, filtlong removed a small number of lower-quality reads but did not significantly change the overall read length distribution.

ONT

	Before	After
Number of reads	175,250	119,889
Total bases (Gb)	3.24	3.08
Mean read length (bp)	18,486.1	25,671.4
Median read length (bp)	6,864.5	12,849.0
Read length N50 (bp)	51,341	53,333
Mean read quality	9.0	9.7
Median read quality	10.2	10.7
Reads > Q10 (%)	52.9	61.3

- The number of reads decreases ($175,250 \rightarrow 119,889$), while total bases decrease only slightly ($3.24 \text{ Gb} \rightarrow 3.08 \text{ Gb}$), indicating removal of shorter reads. Mean and median read length increase markedly ($18.5 \text{ kb} \rightarrow 25.7 \text{ kb}$; $6.9 \text{ kb} \rightarrow 12.8 \text{ kb}$), and read quality improves modestly (mean Q $9.0 \rightarrow 9.7$; median Q $10.2 \rightarrow 10.7$). The proportion of reads above Q10 increases from 52.9% to 61.3% , and N50 increases slightly ($51.3 \text{ kb} \rightarrow 53.3 \text{ kb}$).
- From the plots, the read length versus quality plot shows fewer short, low-quality reads after filtering. The read length histograms (including the log-transformed ones) show a reduction in the short-read tail and a general shift toward longer reads. The weighted histograms suggest that longer reads now make up a larger proportion of the total bases. The yield-by-length plots show a slight decrease in total

yield. This reflects that filtering did not alter the overall structure of the dataset but only removed a subset of low-quality reads.

Task 5: MiniMapping - Create a slurm job.

illumina - Mapping

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
#SBATCH --time=04:00:00
#SBATCH --job-name=map_short_reads
#SBATCH --array=1
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_1/logs/map_%A_%a.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_1/logs/map_%A_%a.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/1_seq_techs/conda_env/1_seq_techs

CHR=chr20

REF=/project2/biodb/genomes/Homo_sapiens/NCBI/GRCh38/Sequence/BWAIndex/genome.fa
IN=/project2/msalomon_1816/ranzhu/assignment_1/fastq
OUT=/project2/msalomon_1816/ranzhu/assignment_1/mapping
mkdir -p $OUT

SAMPLE=hg200

flowcell=$(zcat $IN/hg200_illumina_chr20_1.fastq.gz | head -n1 | cut -d':' -f3)
lane=$(zcat $IN/hg200_illumina_chr20_1.fastq.gz | head -n1 | cut -d':' -f4)

rg_id=${flowcell}.${lane}
pu=${flowcell}.${lane}.${SAMPLE}
pl=ILLUMINA

bwa mem -t 4 \
  -R "@RG\tID:${rg_id}\tPU:${pu}\tPL:${pl}\tSM:${SAMPLE}" \
  $REF \
  $IN/hg200_illumina_chr20_1.fastq.gz \
  $IN/hg200_illumina_chr20_2.fastq.gz \
  | samtools view -bS - \
  | samtools sort -@ 4 -T $OUT/temp -O bam -o $OUT/illumina.chr20.all.bam

samtools index $OUT/illumina.chr20.all.bam
samtools view -b $OUT/illumina.chr20.all.bam $CHR > $OUT/illumina.chr20.sorted.bam
samtools index $OUT/illumina.chr20.sorted.bam
rm $OUT/illumina.chr20.all.bam
```

illumina - duplicates

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
#SBATCH --time=02:00:00
#SBATCH --job-name=markdup_samtools
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_1/logs/markdup_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_1/logs/markdup_%j.err

# activate environment
source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/1_seq_techs/conda_env/1_seq_techs

IN=/project2/msalomon_1816/ranzhu/assignment_1/mapping/illumina.chr20.sorted.bam
OUT=/project2/msalomon_1816/ranzhu/assignment_1/mapping

samtools collate -Ou $IN | \
  samtools fixmate -m - -u | \
  samtools sort -u | \
  samtools markdup - $OUT/illumina.chr20.dedup.bam

samtools index $OUT/illumina.chr20.dedup.bam
```

Pacbio - minimapping

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
#SBATCH --time=04:00:00
#SBATCH --job-name=map_pacbio_reads
#SBATCH --array=1
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_1/logs/map_pacbio_%A_%a.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_1/logs/map_pacbio_%A_%a.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/1_seq_techs/conda_env/1_seq_techs

CHR=chr20
REF=/project2/biodb/genomes/Homo_sapiens/NCBI/GRCh38/Sequence/WholeGenomeFasta/genome.fa
IN=/project2/msalomon_1816/ranzhu/assignment_1/fastq
OUT=/project2/msalomon_1816/ranzhu/assignment_1/mapping
```

```

minimap2 -ax map-hifi -t 4 \
  $REF \
  $IN/hg200_pacbio_chr20.fastq.gz \
  | samtools view -bS - \
  | samtools sort -@ 4 -T $OUT/temp_pacbio -O bam -o $OUT/pacbio.chr20.all.bam

samtools index $OUT/pacbio.chr20.all.bam

samtools view -b $OUT/pacbio.chr20.all.bam $CHR > $OUT/pacbio.chr20.sorted.bam

samtools index $OUT/pacbio.chr20.sorted.bam

rm $OUT/pacbio.chr20.all.bam $OUT/pacbio.chr20.all.bam.bai

```

ONT - minimapping

```

#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
#SBATCH --time=04:00:00
#SBATCH --job-name=map_ont_reads
#SBATCH --array=1
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_1/logs/map_ont_%A_%a.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_1/logs/map_ont_%A_%a.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/1_seq_techs/conda_env/1_seq_techs

CHR=chr20
REF=/project2/biodb/genomes/Homo_sapiens/NCBI/GRCh38/Sequence/WholeGenomeFasta/genome.fa
IN=/project2/msalomon_1816/ranzhu/assignment_1/fastq
OUT=/project2/msalomon_1816/ranzhu/assignment_1/mapping

SAMPLE=hg200

minimap2 -ax map-ont -t 4 \
  $REF \
  $IN/hg200_ont_chr20.fastq.gz \
  | samtools view -bS - \
  | samtools sort -@ 4 -T $OUT/temp_ont -O bam -o $OUT/ont.chr20.all.bam

samtools index $OUT/ont.chr20.all.bam

samtools view -b $OUT/ont.chr20.all.bam $CHR > $OUT/ont.chr20.sorted.bam

samtools index $OUT/ont.chr20.sorted.bam

rm $OUT/ont.chr20.all.bam $OUT/ont.chr20.all.bam.bai

```

Task 6: short reads variant calling

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
#SBATCH --time=02:00:00
#SBATCH --job-name=markdup_samtools
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_1/logs/markdup_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_1/logs/markdup_%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/1_seq_techs/conda_env/1_seq_techs

mkdir -p /project2/msalomon_1816/ranzhu/assignment_1/vcf

REF=/project2/msalomon_1816/trgn_515/1_seq_techs/reference_genome/genome.fa
input_dir=/project2/msalomon_1816/ranzhu/assignment_1/mapping/
out_dir=/project2/msalomon_1816/ranzhu/assignment_1/vcf
input_bam=$input_dir/illumina.chr20.sorted.bam

bcftools mpileup \
-f $REF \
-a AD,INFO/AD,ADF,INFO/ADF,ADR,INFO/ADR,DP,SP \
-I \
-q 0 \
-Q 0 \
-Ou \
$input_bam \
|bcftools call \
-m \
-v \
-V indels \
-Oz \
|bcftools norm \
-f $REF \
-Oz \
-o $out_dir/illumina.output.vcf.gz
```

filter_illumina

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
```

```

#SBATCH --time=02:00:00
#SBATCH --job-name=markdup_samtools
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_1/logs/markdup_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_1/logs/markdup_%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/1_seq_techs/conda_env/1_seq_techs

vcf_dir=/project2/msalomon_1816/ranzhu/assignment_1/vcf
input_vcf=$vcf_dir/illumina.output.vcf.gz
output_vcf=$vcf_dir/illumina.task6.filtered.vcf.gz

# Soft-filter according to Task 6

bcftools filter -m+ -s 'MinMQ' -e 'INFO/MQ < 20' $input_vcf \
| bcftools filter -m+ -s 'QUAL' -e 'QUAL < 20' \
| bcftools filter -m+ -s 'minAD' -e 'FMT/AD[:1] < 5' \
| bcftools filter -m+ -s 'minADF' -e 'FMT/ADF[:1] < 5' \
| bcftools filter -m+ -s 'minADR' -e 'FMT/ADR[:1] < 5' \
| bcftools filter -m+ -s 'MinDP' -e 'INFO/DP < 20' \
-Oz -o $output_vcf

tabix -p vcf $output_vcf

```

Task 7: variant_calling

illumina

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
#SBATCH --time=02:00:00
#SBATCH --job-name=task7_illumina
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_1/logs/task7_illumina_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_1/logs/task7_illumina_%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/1_seq_techs/conda_env/1_seq_techs

REF=/project2/msalomon_1816/trgn_515/1_seq_techs/reference_genome/genome.fa
input_dir=/project2/msalomon_1816/ranzhu/assignment_1/mapping
out_dir=/project2/msalomon_1816/ranzhu/assignment_1/vcf
input_bam=$input_dir/illumina.chr20.sorted.bam

bcftools mpileup \
-f $REF \
-a AD,INFO/AD,ADF,INFO/ADF,ADR,INFO/ADR,DP \
-q 0 \
-Q 20 \
-Ou \
$input_bam \
| bcftools call \
-m \
-v \
-V indels \
-Ou \
| bcftools norm \
-f $REF \
-Ou \
| bcftools filter -m+ -s 'MinMQ' -e 'INFO/MQ < 10' \
| bcftools filter -m+ -s 'QUAL' -e 'QUAL < 10' \
| bcftools filter -m+ -s 'MinDP' -e 'INFO/DP < 20' \
| bcftools filter -m+ -s 'minAD' -e 'FMT/AD[:1] < 5' \
| bcftools filter -m+ -s 'minADF' -e 'FMT/ADF[:1] < 2' \
| bcftools filter -m+ -s 'minADR' -e 'FMT/ADR[:1] < 2' \
-Oz -o $out_dir/illumina.task7.filtered.vcf.gz

tabix -p vcf $out_dir/illumina.task7.filtered.vcf.gz
```

Pacbio

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
#SBATCH --time=02:00:00
#SBATCH --job-name=task7_pacbio
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_1/logs/task7_pacbio_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_1/logs/task7_pacbio_%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/1_seq_techs/conda_env/1_seq_techs

REF=/project2/msalomon_1816/trgn_515/1_seq_techs/reference_genome/genome.fa
input_dir=/project2/msalomon_1816/ranzhu/assignment_1/mapping
out_dir=/project2/msalomon_1816/ranzhu/assignment_1/vcf
input_bam=$input_dir/pacbio.chr20.sorted.bam

bcftools mpileup \
-f $REF \
-a AD,INFO/AD,DP \
-q 0 \
-Q 20 \
-Ou \
$input_bam \
| bcftools call \
-m \
-v \
-V indels \
-Ou \
| bcftools norm \
-f $REF \
-Ou \
| bcftools filter -m+ -s 'MinMQ' -e 'INFO/MQ < 10' \
| bcftools filter -m+ -s 'QUAL' -e 'QUAL < 10' \
| bcftools filter -m+ -s 'MinDP' -e 'INFO/DP < 20' \
| bcftools filter -m+ -s 'minAD' -e 'FMT/AD[:1] < 5' \
| bcftools filter -m+ -s 'minADR' -e 'FMT/ADR[:1] < 2' \
| bcftools filter -m+ -s 'minADF' -e 'FMT/ADF[:1] < 2' \
-Oz -o $out_dir/pacbio.task7.filtered.vcf.gz

tabix -p vcf $out_dir/pacbio.task7.filtered.vcf.gz
```

ONT

```
#!/bin/bash
#SBATCH --account=msalomon_1816
```

```

#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=48G
#SBATCH --time=48:00:00
#SBATCH --job-name=task7_ont
#SBATCH --output=/project2/msalomon_1816/ranzhu/assignment_1/logs/task7_ont_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/assignment_1/logs/task7_ont_%j.err
#SBATCH --mail-type=ALL
#SBATCH --mail-user=ranzhu@usc.edu

set -euo pipefail

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/1_seq_techs/conda_env/1_seq_techs

REF=/project2/msalomon_1816/trgn_515/1_seq_techs/reference_genome/genome.fa
input_bam=/project2/msalomon_1816/ranzhu/assignment_1/mapping/ont.chr20.sorted.bam
out_dir=/project2/msalomon_1816/ranzhu/assignment_1/vcf
mkdir -p $out_dir

bcftools mpileup \
--threads 8 \
-f $REF \
-a AD,ADF,ADR,DP \
-q 0 \
-Q 20 \
-I \
-Ou \
$input_bam \
| bcftools call \
--threads 8 \
-m \
-v \
-V indels \
-Ou \
| bcftools norm \
--threads 8 \
-f $REF \
-Ou \
| bcftools filter -m+ -s 'MinMQ' -e 'INFO/MQ < 5' -Ou \
| bcftools filter -m+ -s 'QUAL' -e 'QUAL < 5' -Ou \
| bcftools filter -m+ -s 'MinDP' -e 'INFO/DP < 20' -Ou \
| bcftools filter -m+ -s 'minAD' -e 'SUM(FMT/AD[0:1]) < 5' -Ou \
| bcftools filter -m+ -s 'minADR' -e 'SUM(FMT/ADR[0:1]) < 2' -Ou \
| bcftools filter -m+ -s 'minADF' -e 'SUM(FMT/ADF[0:1]) < 2' \
-Oz -o $out_dir/ont.task7.filtered.vcf.gz

tabix -p vcf $out_dir/ont.task7.filtered.vcf.gz

```

Question 8: Benchmarking

standard

VCF intersection	Benchmark metric
Va and Vb	True Positive (TP)
not Va and not Vb	True Negative (TN)
Va and not Vb	False Positive (FP)
not Va and Vb	False Negative (FN)

illumina

File	illumina	Ben	Metric
0000.vcf	Yes	NO	FP
0001.vcf	NO	Yes	FN
0002.vcf	Yes	Yes	TP
0003.vcf	Yes	Yes	

Illumina:

- $TP = 3,790$
- $FP = 1,897$
- $FN = 69,599$
- $Precision = TP / (TP + FP) = 0.666$
- $Recall = TP / (TP + FN) = 0.0516$
- $F1 = 2 \times 0.666 \times 0.0516 / (0.666 + 0.0516) = 0.096$

pacbio

File	Pacbio	Ben	Metric
0000.vcf	Yes	NO	FP
0001.vcf	NO	Yes	FN
0002.vcf	Yes	Yes	TP
0003.vcf	Yes	Yes	

Pacbio:

- $TP = 73,247$
- $FP = 80,253$
- $FN = 142$
- $Precision = 73247 / (73247 + 80253) = 73247 / 153500 = 0.477$
- $Recall = 73247 / (73247 + 142) = 73247 / 73389 = 0.998$
- $F1 = 0.64$

ONT

File	ONT	Ben	Metric
0000.vcf	Yes	NO	FP
0001.vcf	NO	Yes	FN
0002.vcf	Yes	Yes	TP
0003.vcf	Yes	Yes	

ONT:

- $TP = 70,571$
- $FP = 205,333$
- $FN = 2,818$
- $Precision = 70,571/275,904 = 0.256$
- $Recall = 70,571/73,389 = 0.962$
- $F1 = 0.41$

Question 9: Genome-wide assessment using genomic windows

1. How many variants are called using each sequencing technology and how many are in the gold standard? (5pts)

- $V_a = TP + FP$; $V_b = TP + FN$
- gold_chr20 = 73,389 variants
- ONT total calls = $TP + FP = 70,571 + 205,333 = 275,904$ variants
- PacBio total calls = $73,247 + 80,253 = 153,500$ variants
- Illumina total calls = $3,790 + 1,897 = 5,687$ variants

2. How many true positive, false positive and false negative variants are present from using the different sequencing technology? (5pts)

- ONT: 70,571 TP, 205,333 FP, 2,818 FN
- PacBio: 73,247 TP, 80,253 FP, 142 FN
- Illumina: 3,790 TP, 1,897 FP, 69,599 FN

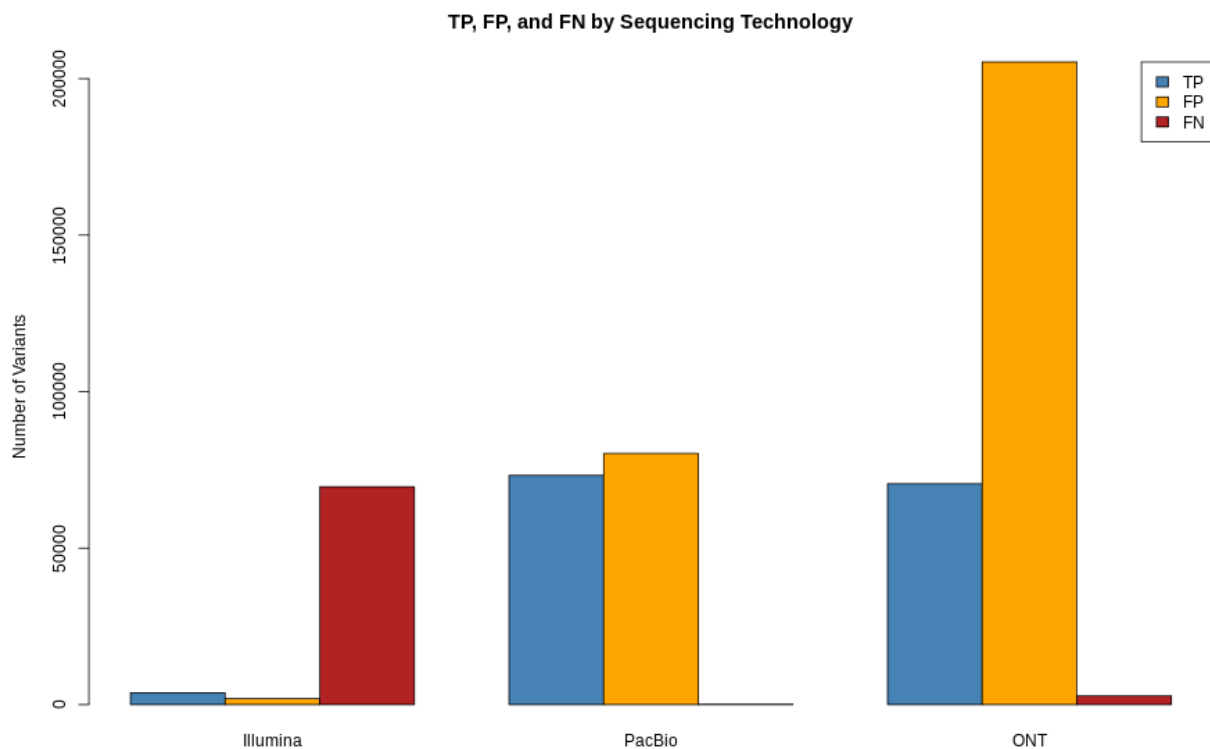
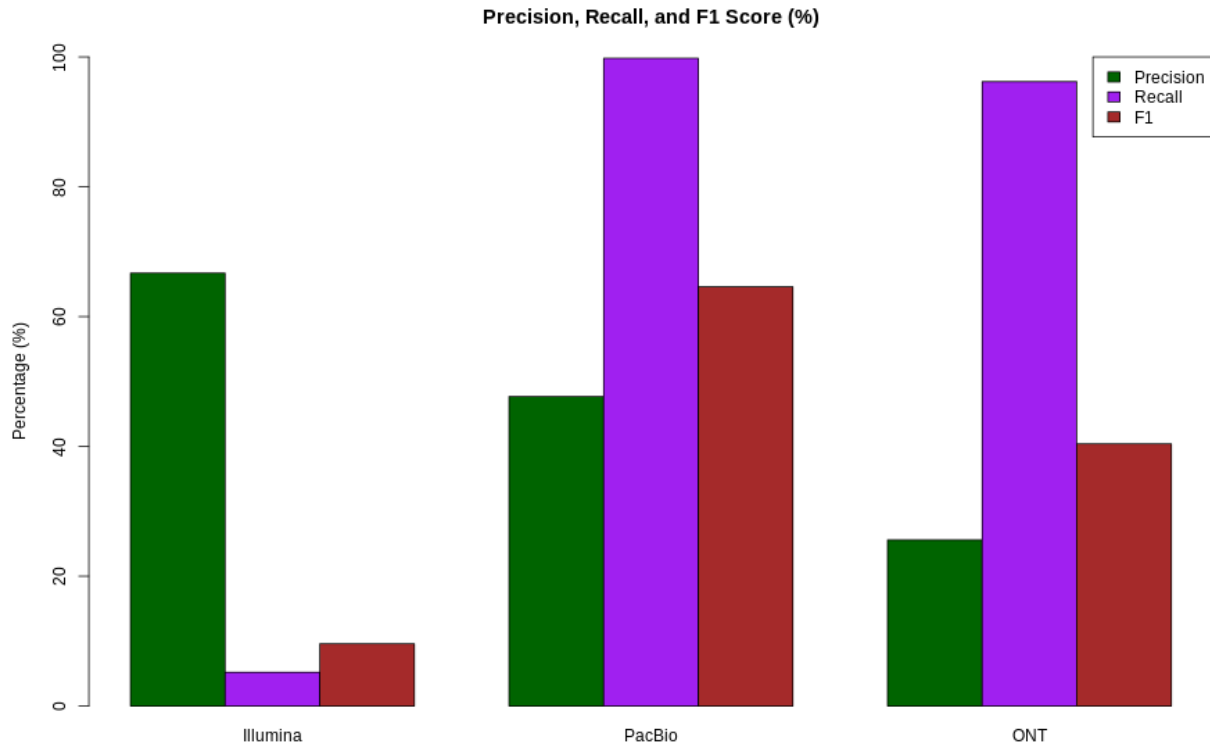
3. Calculate the precision and recall for each sequencing technology (5pts)

- Precision = $TP / (TP + FP)$
- ONT: $70571 / 275904 = 0.256$
- Pacbio: $73247 / 153500 = 0.477$
- Illumina: $3790 / (3790 + 1896) = 0.67$
- Recall = $TP / (TP + FN)$
- ONT: $70571 / 73389 = 0.962$
- Pacbio: $73247 / 73389 = 0.99$
- illumina: $3790 / 73389 = 0.052$
- ONT achieved a precision of 25.6% and a recall of 96.2%.
- PacBio achieved a precision of 47.7% and the highest recall 99%.
- Illumina achieved a precision of 67% and low recall 5.2%.

4. Calculate the F1 score for each sequencing technology (5pts)

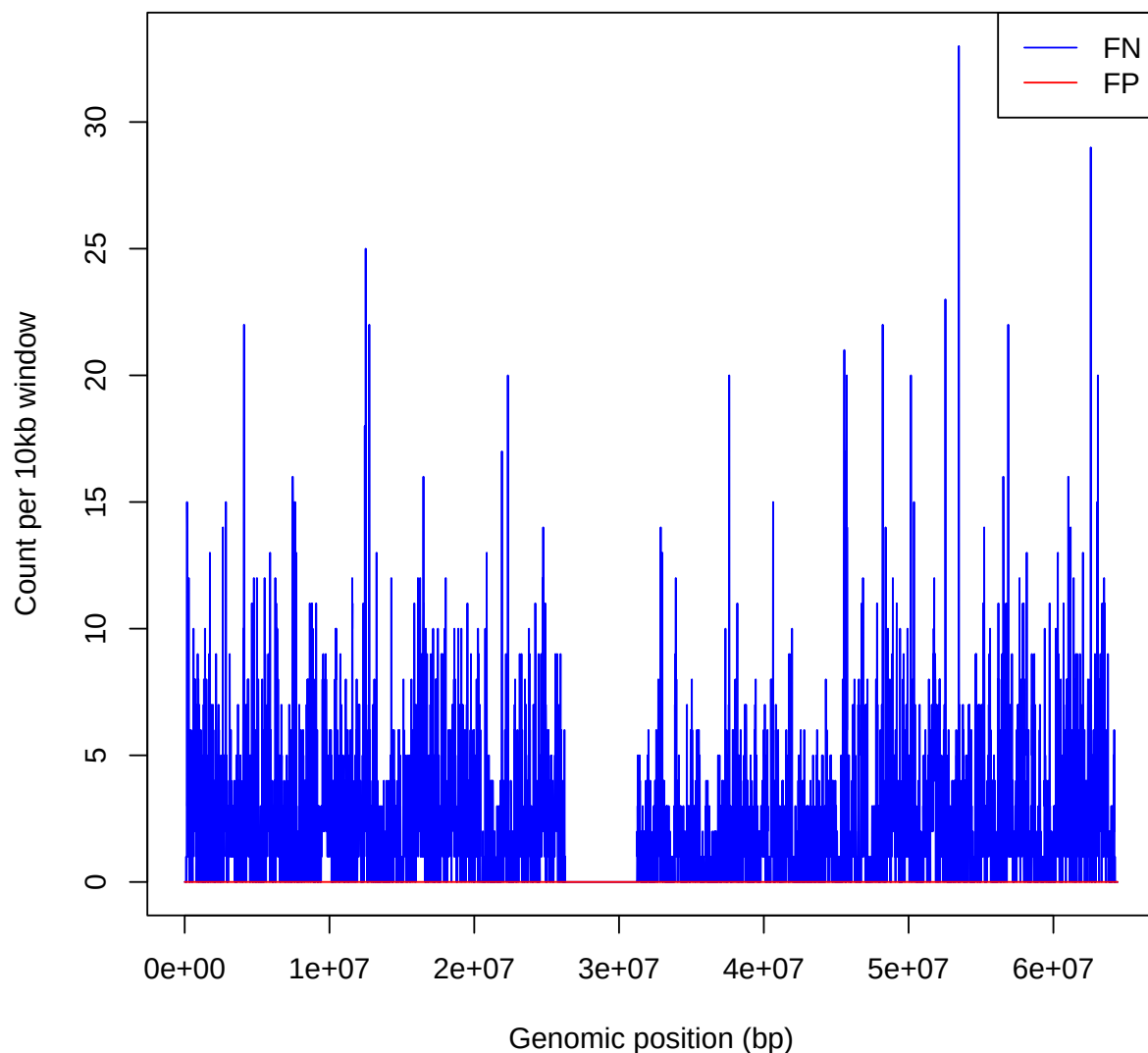
- $F1 = (2 \times \text{Precision} \times \text{Recall}) / (\text{precision} + \text{recall})$
- ONT $F1 = (2 \times 0.25 \times 0.96) / (0.25 + 0.96) = 0.4$
- Pacbio $F1 = (2 \times 0.47 \times 0.99) / (0.47 + 0.99) = 0.65$
- Illumina $F1 = (2 \times 0.67 \times 0.05) / (0.67 + 0.05) = 0.096$

5. Make a grouped barplot of each metric and each sequencing technology. Group bars by sequencing technology. Make one plot for TP, FP, FN; and another plot for precision, recall and F1 score (in percentage) (10pts)

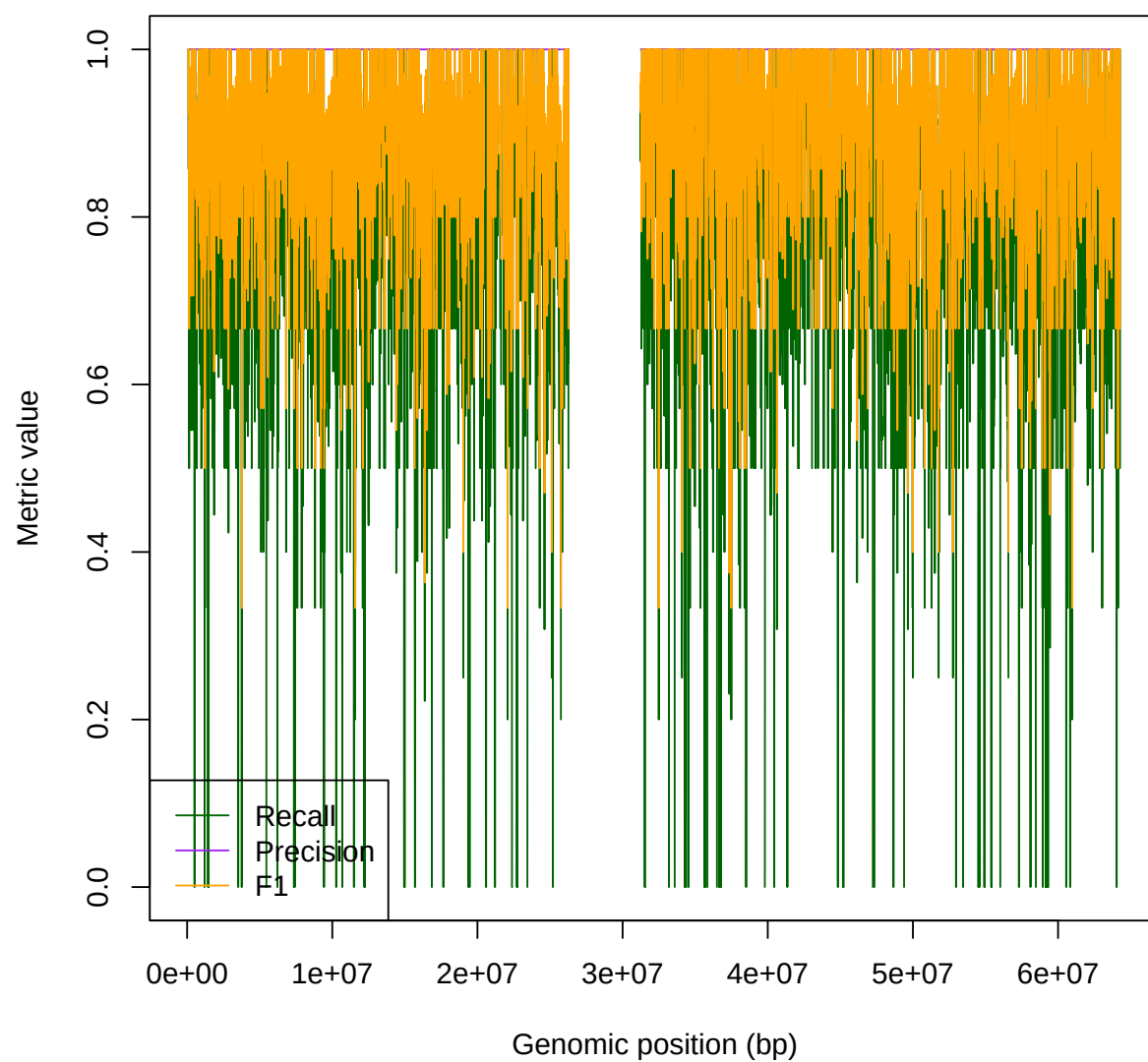


6. Look at the distribution of false negatives and false positives along the genome per sequencing technology using a genomic window of 10kbp. Make a density plot or a histogram of FP and FN along the genome. X-axis is genome position as determined by the beginning of the window. Y-axis is the number of FP or FN. Is there any pattern in the distribution? Are there any areas with high FP and FN? Calculate recall, precision and F1 for each genomic window. Are there any areas with low recall/precision/F1? Give approximate genomic coordinates (20pts)

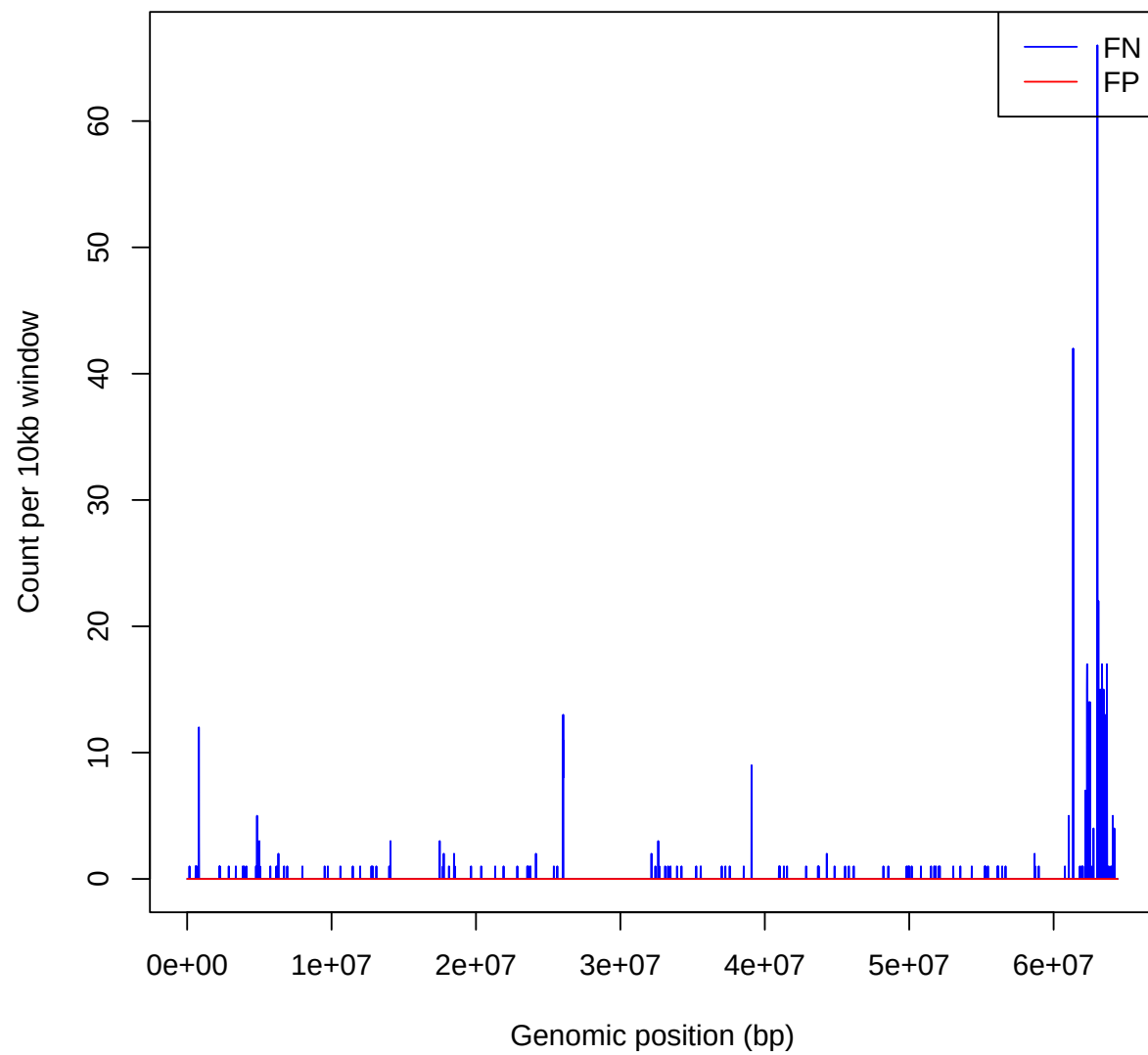
ONT FP and FN distribution (chr20, 10kb)



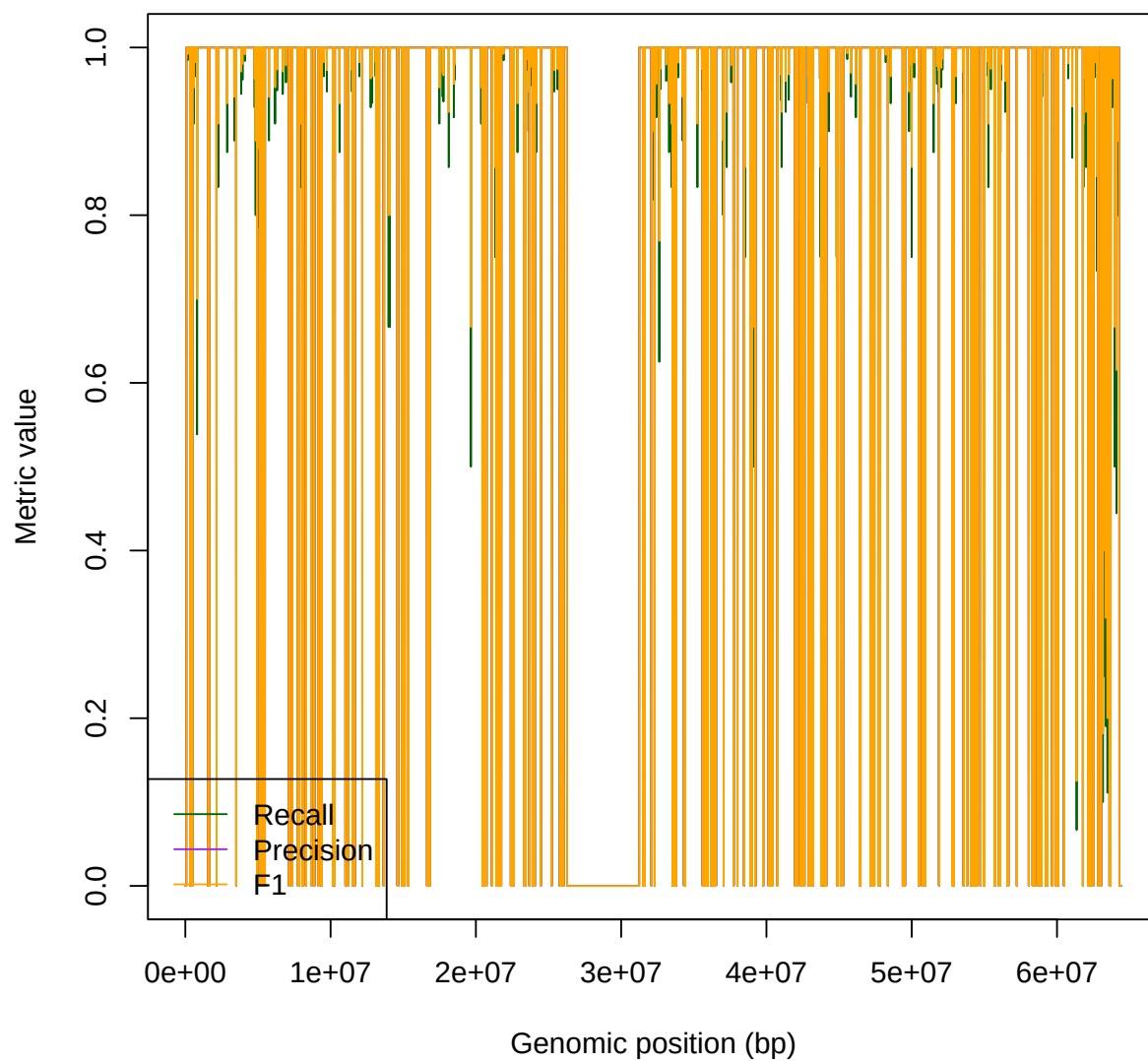
ONT Precision, Recall and F1 (10kb windows)



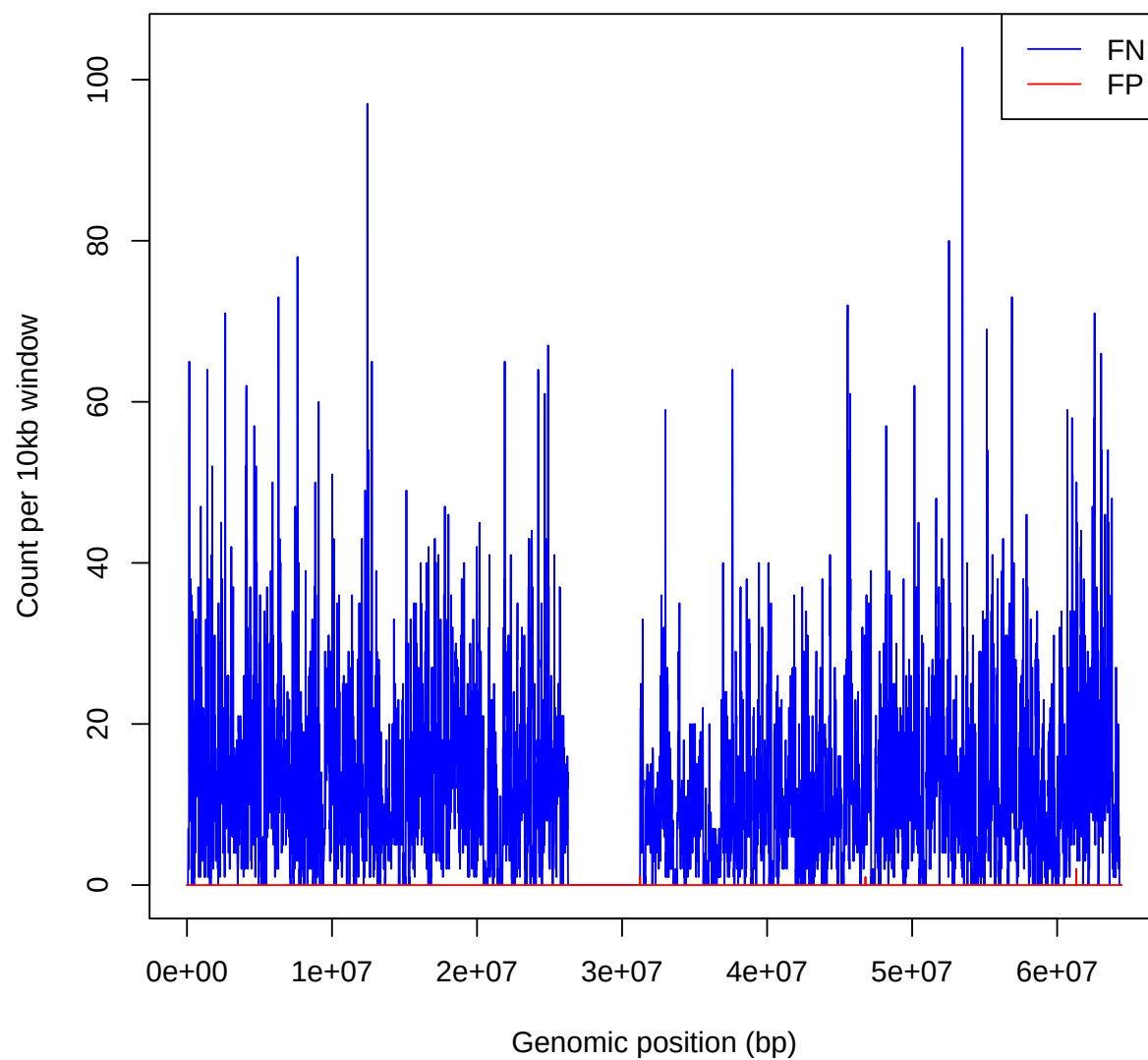
PacBio FP and FN distribution (10kb)



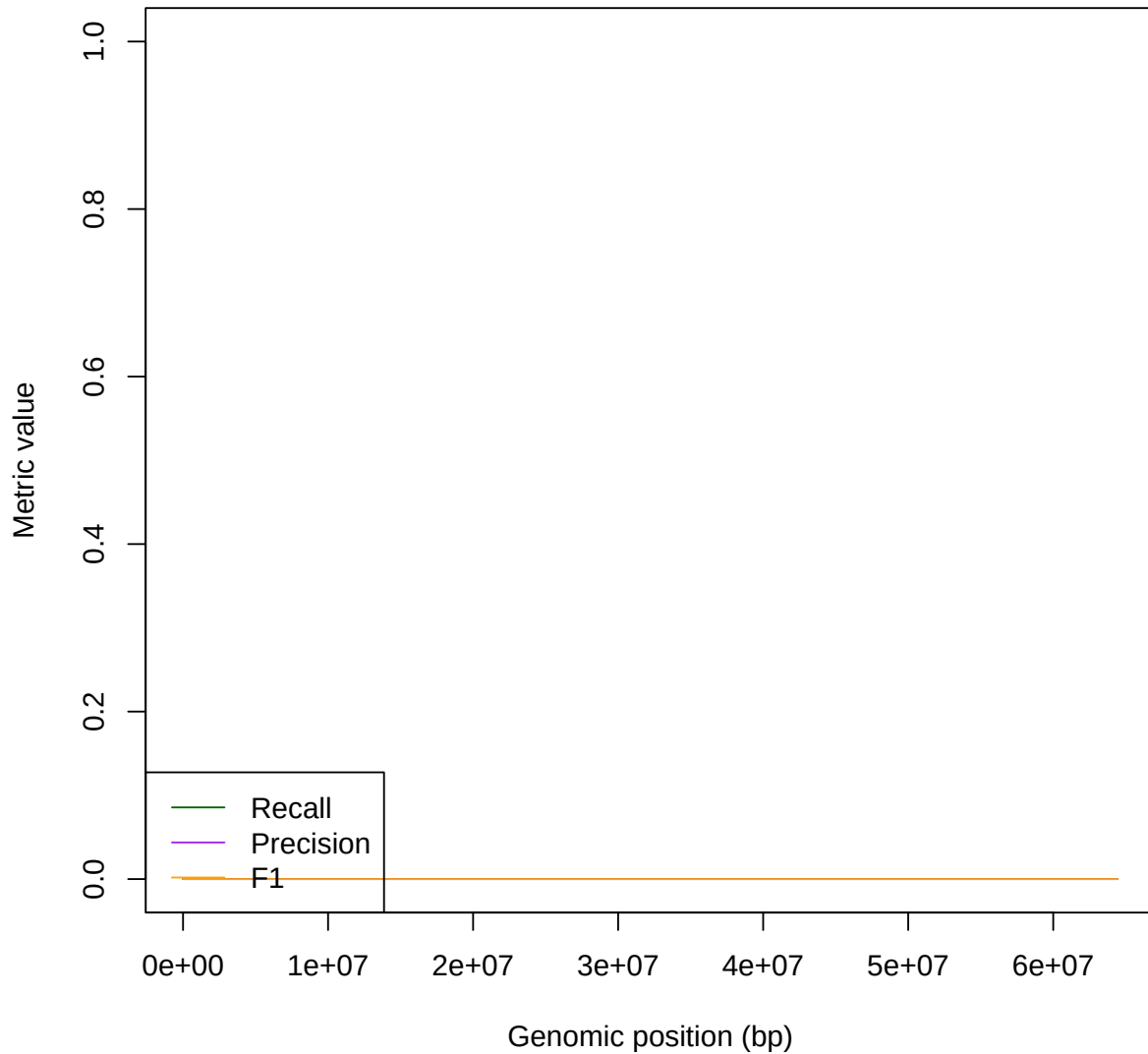
PacBio Precision, Recall and F1 (10kb)



Illumina FP and FN distribution (10kb)



Illumina Precision, Recall and F1 (10kb)



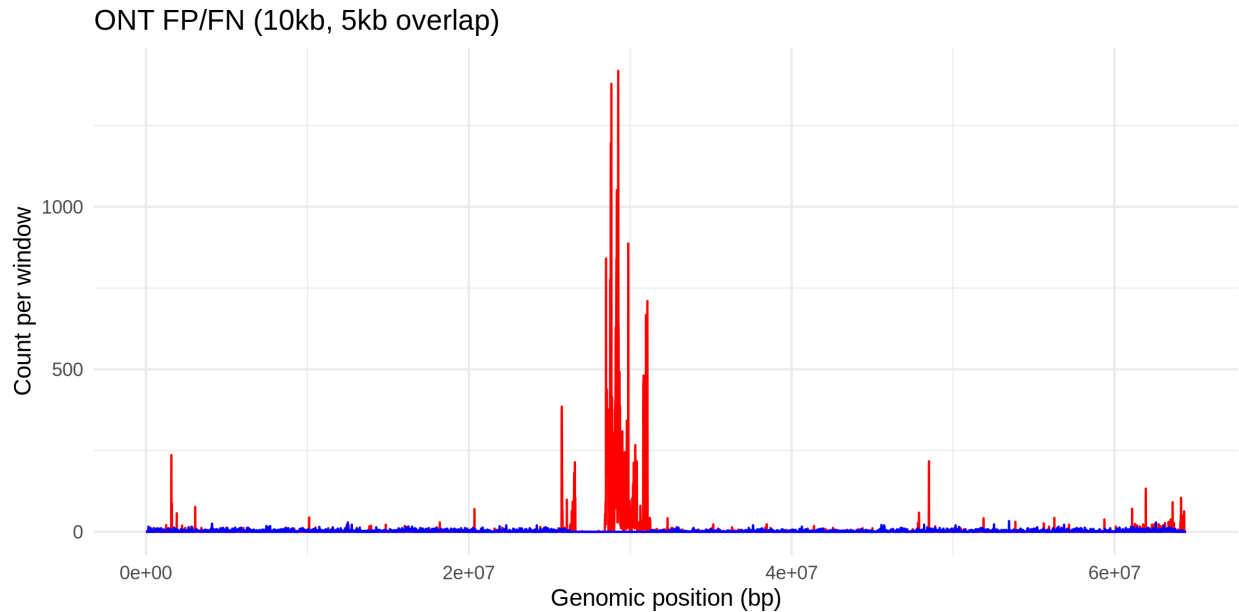
FP and FN Distribution

- ONT: FN counts are spread across the whole chromosome, around 5–15 per window, with the highest peaks around ~15 Mb, ~20 Mb, and ~63 Mb. FP counts are not visible in the plot, appearing near zero in every window. The gap around ~26–30 Mb corresponds to the centromeric region.
- Pacbio: FN counts are mostly near 0 across the majority of chr20, with only a few isolated small peaks. There is a small peak at ~1 Mb, a few small peaks around ~5 Mb, ~18 Mb, and ~25 Mb. After the centromeric gap around ~26–30 Mb, FN counts remain very low until near the end of the chromosome, where there is a high peak at ~62 Mb reaching ~42 and another at ~63 Mb reaching ~60. FP counts stay near zero throughout the entire chromosome.
- illumina: The centromeric gap is visible around ~26–30 Mb where FN drops close to zero. FP counts stay near zero across the entire chromosome. Overall the FN distribution is much higher and more uniform compared to PacBio and ONT, reflecting the very low global recall of 0.052.

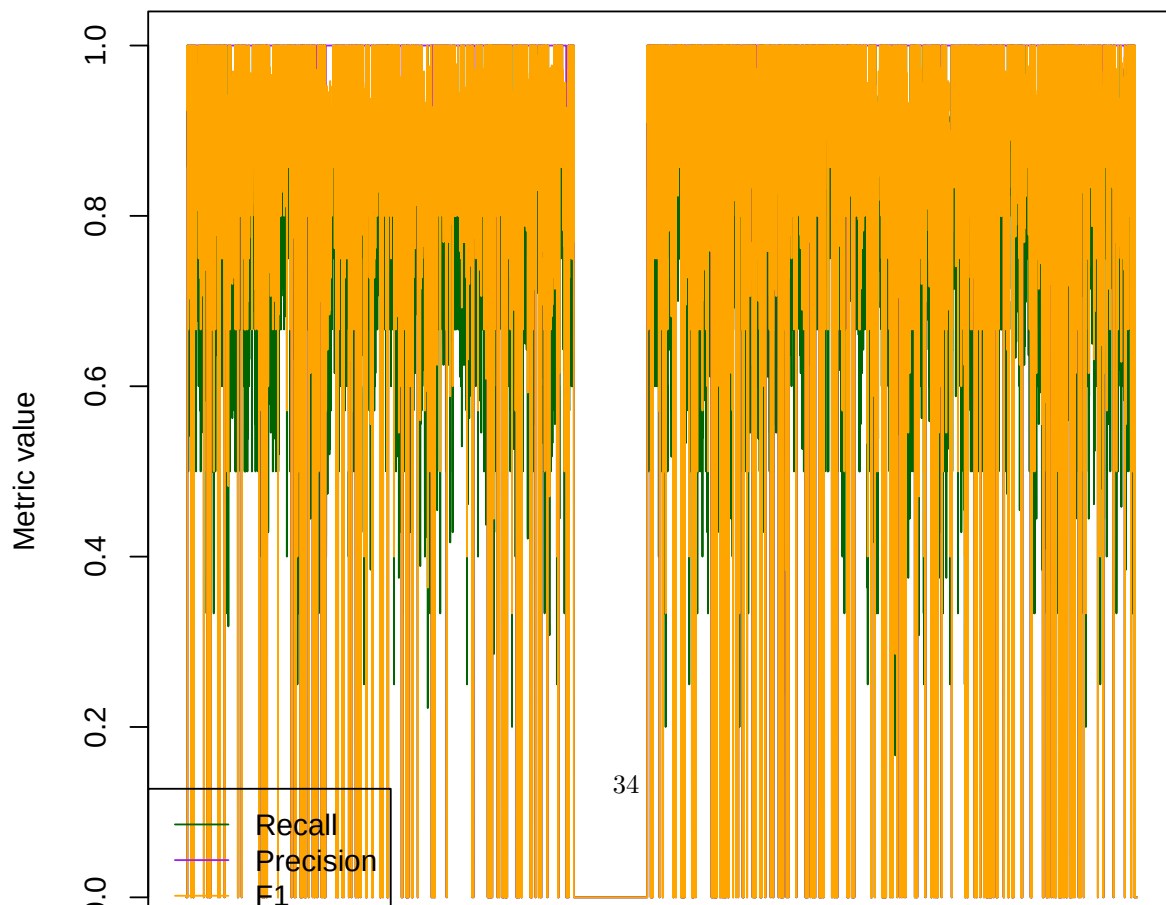
Recall

- ONT: Recall and F1 are generally around 0.6–0.8 but drop to around 0.2–0.4 in some windows throughout the chromosome, with the lowest areas around ~15–20 Mb and ~60–64 Mb. Precision stays close to 1.0 in most windows.
- Pacbio: Precision stays near 1.0 across the entire chromosome. Recall is mostly close to 1.0 for most windows, but drops frequently throughout the chromosome, going down to around 0.6–0.8 in many windows and occasionally dropping all the way to 0. The low recall region is around ~62–64 Mb, where recall drops to 0 in several consecutive windows, matching the large FN peak in that region. F1 follows the same pattern as recall since precision stays high throughout. The centromeric gap is visible around ~26–30 Mb.
- illumina: Recall, precision and F1 are 0 across the entire chromosome. All three lines stay flat at the bottom of the plot with no variation. This aligns well with recall calculation of 0.052 and the fact that almost no TP were detected per window due to the strict filtering applied in Variant calling steps.

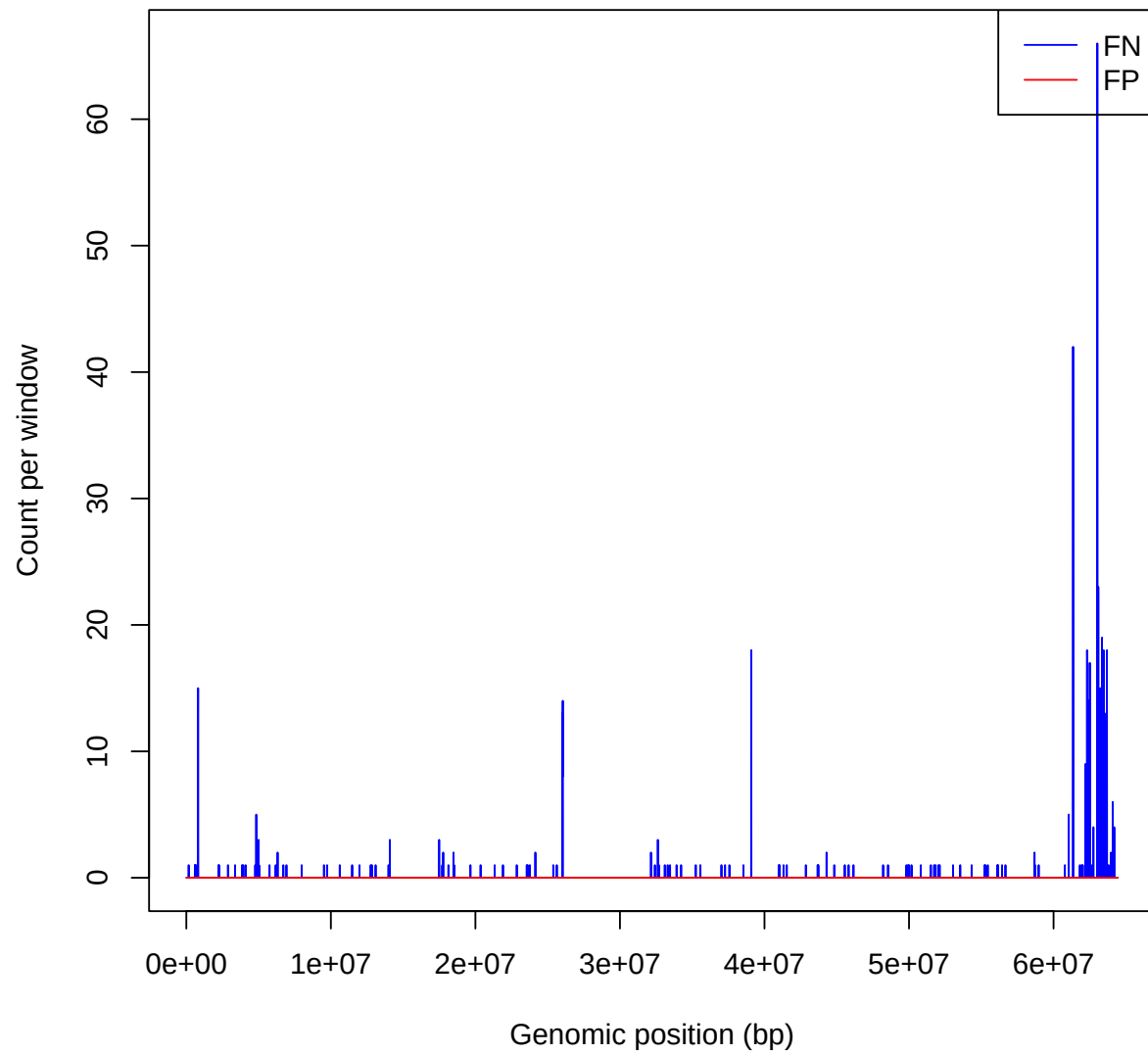
7. Repeat the analysis using genomic windows of 10kbp and a 5000bp window overlap. You can find this in the bedtools makewindows command help. Repeat the plot of the previous question. How has the plot change? Is there any pattern in the distribution? Are there any areas with high FP and FN? Calculate recall, precision and F1 for each genomic window. Are there any areas with low recall/precision/F1? Give approximate genomic coordinates (20pts)



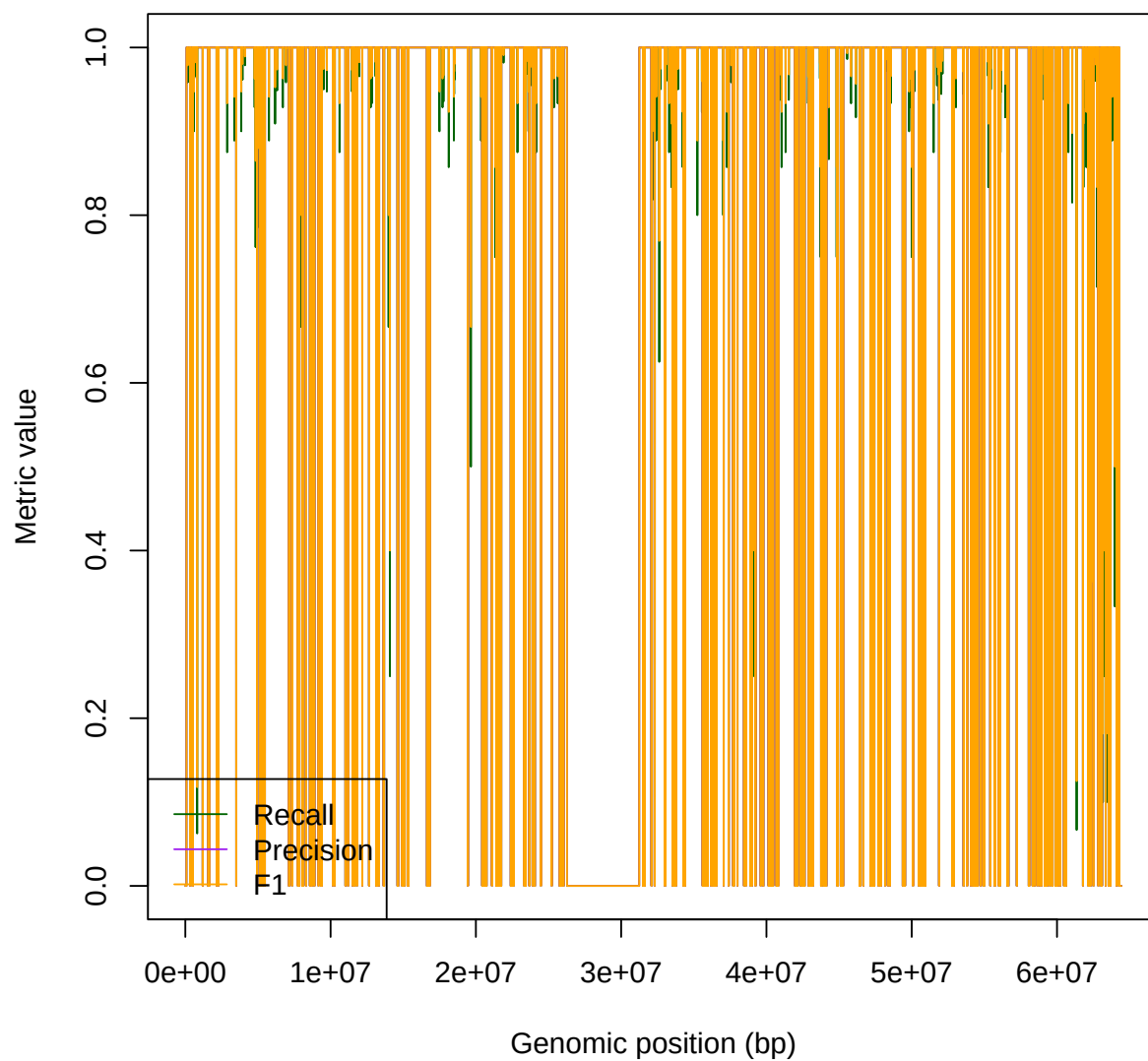
ONT Precision, Recall and F1 (10kb, 5kb overlap)



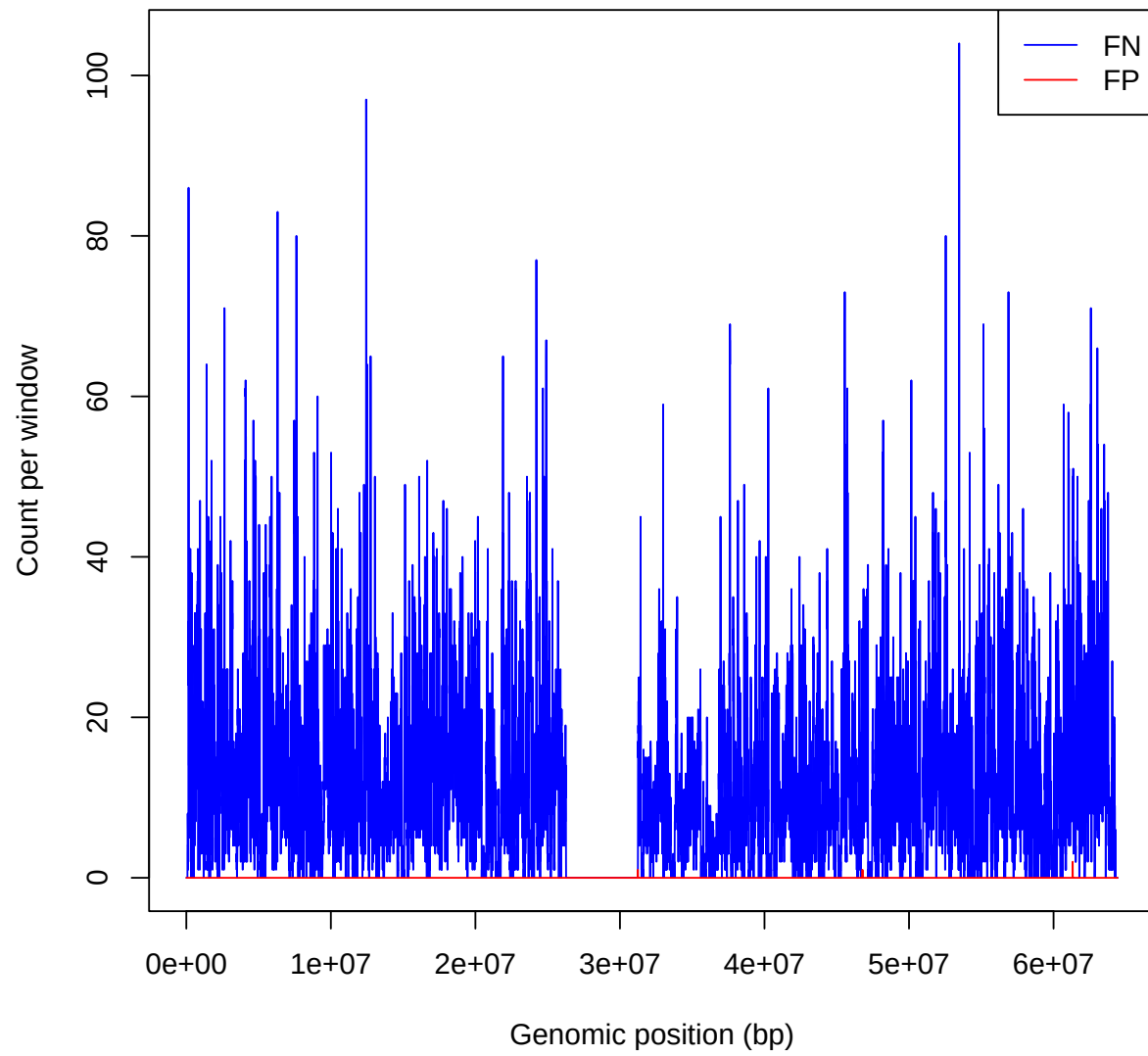
PacBio FP and FN (10kb, 5kb overlap)



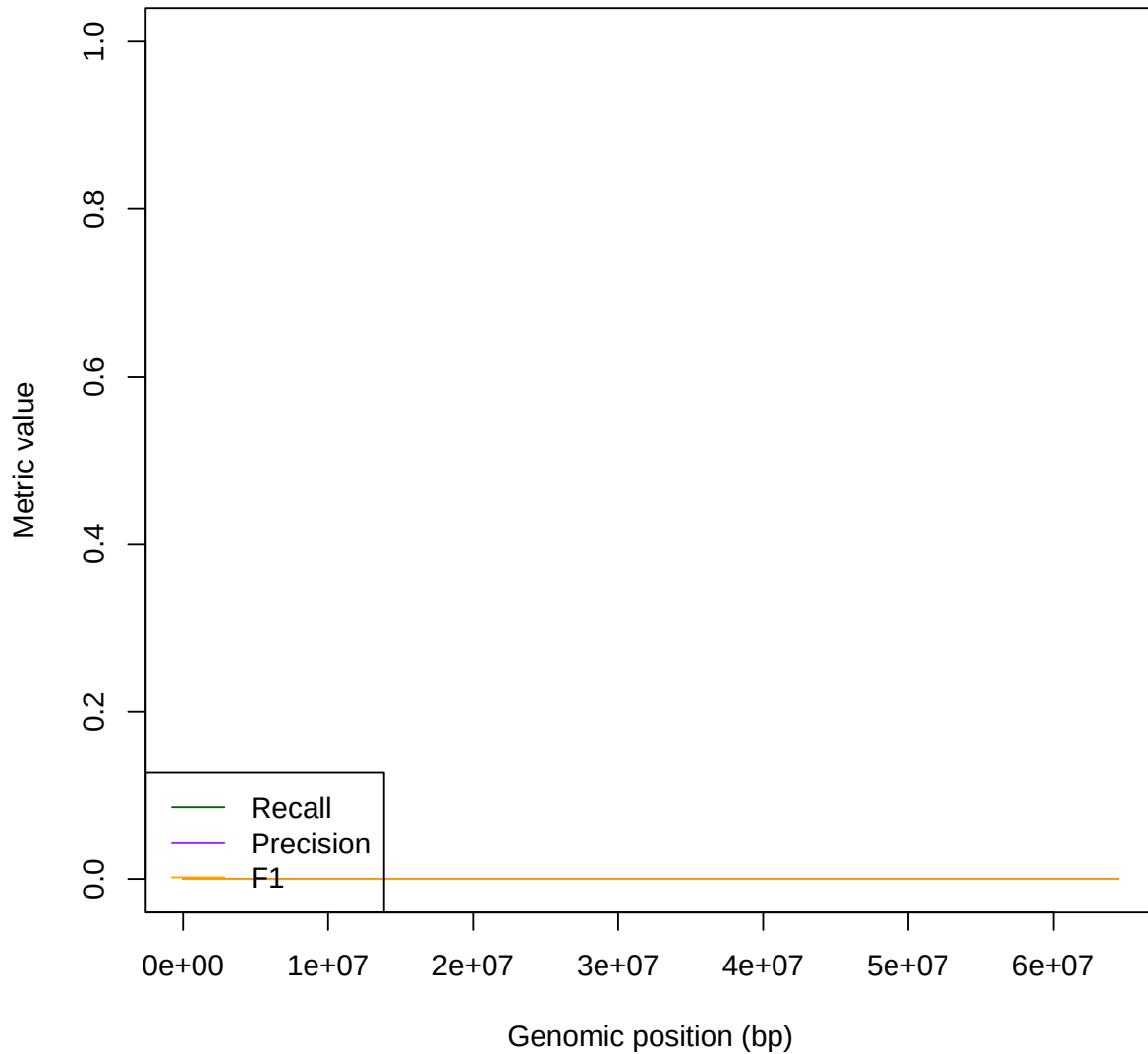
PacBio Precision, Recall and F1 (overlap)



Illumina FP and FN (10kb, 5kb overlap)



Illumina Precision, Recall and F1 (overlap)



- ONT: Compared to the non-overlapping 10kb windows, FN counts remain low across the chromosome, similar to before. FP counts now show a highest peak reaching over 1,200 per window. There are also smaller FP spikes around ~22 Mb and ~60–63 Mb. Recall is generally around 0.6–0.8 as before, with drops to around 0.2 in scattered windows. F1 follows recall for most of the chromosome but also drops in the ~28–35 Mb region due to the precision drop there. This is a new pattern not visible in the non-overlapping plot.
- Pacbio: Compared to the non-overlapping plot, the overall pattern is very similar. FN counts are mostly near 0 across the chromosome, and FP stays near zero throughout. Precision stays near 1.0 across the whole chromosome. Recall is mostly close to 1.0 but drops frequently to 0 in scattered windows throughout. The drops are more visible now due to overlapping windows, with notable dips around ~5 Mb, ~13 Mb, and more frequently near ~62–64 Mb where recall drops to 0 in several consecutive windows. F1 follows the same pattern as recall.
- Illumina: FN counts remain high genome-wide with a baseline of around 20–40 per window. The centromeric gap still around ~26–30 Mb is clearly visible. For recall plot, same as the non-overlapping

plot. All three metrics stay flat at 0 across the entire chromosome.